Appendix

0.1 Importing the Required Libraries

[34]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
import scipy.stats as stats
from scipy.stats import pearsonr, f_oneway, ttest_ind, chi2_contingency
from IPython.display import display
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.formula.api import ols
from statsmodels.stats.diagnostic import het_breuschpagan
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import Lasso, Ridge
from sklearn.metrics import mean_squared_error, r2_score
```

0.2 Inspecting the Dataset

We begin by loading the dataset and examining the first few rows:

[35]:
```
# Load the dataset
df_raw = pd.read_csv('../data/student_performance.csv')

# Display first few rows and summary statistics of numeric columns
pd.set_option('display.max_columns', None)
display(df_raw.head())
display(df_raw.describe())

# Print columns with null values and their counts
null_counts = df_raw.isnull().sum()
print(null_counts[null_counts > 0] if not null_counts[null_counts > 0].empty else 'Dataframe␣
↪has no null values')
print(f'columns: {len(df_raw.columns)}, rows: {len(df_raw)}')
```

```
  school sex  age address famsize Pstatus  Medu  Fedu     Mjob      Fjob  \
0     GP   F   18       U     GT3       A     4     4  at_home   teacher
1     GP   F   17       U     GT3       T     1     1  at_home     other
2     GP   F   15       U     LE3       T     1     1  at_home     other
3     GP   F   15       U     GT3       T     4     2   health  services
4     GP   F   16       U     GT3       T     3     3    other     other

   reason guardian  traveltime  studytime  failures schoolsup famsup paid  \
0  course   mother           2          2         0       yes     no   no
```

```
1  course   father           1          2         0        no    yes   no
2   other   mother           1          2         0       yes     no   no
3    home   mother           1          3         0        no    yes   no
4    home   father           1          2         0        no    yes   no

  activities nursery higher internet romantic  famrel  freetime  goout  Dalc  \
0         no     yes    yes       no       no       4         3      4     1
1         no      no    yes      yes       no       5         3      3     1
2         no     yes    yes      yes       no       4         3      2     2
3        yes     yes    yes      yes      yes       3         2      2     1
4         no     yes    yes       no       no       4         3      2     1

   Walc  health  absences  G1  G2  G3
0     1       3         4   0  11  11
1     1       3         2   9  11  11
2     3       3         6  12  13  12
3     1       5         0  14  14  14
4     2       5         0  11  13  13

              age        Medu        Fedu  traveltime   studytime    failures  \
count  649.000000  649.000000  649.000000  649.000000  649.000000  649.000000
mean    16.744222    2.514638    2.306626    1.568567    1.930663    0.221880
std      1.218138    1.134552    1.099931    0.748660    0.829510    0.593235
min     15.000000    0.000000    0.000000    1.000000    1.000000    0.000000
25%     16.000000    2.000000    1.000000    1.000000    1.000000    0.000000
50%     17.000000    2.000000    2.000000    1.000000    2.000000    0.000000
75%     18.000000    4.000000    3.000000    2.000000    2.000000    0.000000
max     22.000000    4.000000    4.000000    4.000000    4.000000    3.000000

           famrel    freetime       goout        Dalc        Walc      health  \
count  649.000000  649.000000  649.000000  649.000000  649.000000  649.000000
mean     3.930663    3.180277    3.184900    1.502311    2.280431    3.536210
std      0.955717    1.051093    1.175766    0.924834    1.284380    1.446259
min      1.000000    1.000000    1.000000    1.000000    1.000000    1.000000
25%      4.000000    3.000000    2.000000    1.000000    1.000000    2.000000
50%      4.000000    3.000000    3.000000    1.000000    2.000000    4.000000
75%      5.000000    4.000000    4.000000    2.000000    3.000000    5.000000
max      5.000000    5.000000    5.000000    5.000000    5.000000    5.000000

         absences          G1          G2          G3
count  649.000000  649.000000  649.000000  649.000000
mean     3.659476   11.399076   11.570108   11.906009
std      4.640759    2.745265    2.913639    3.230656
min      0.000000    0.000000    0.000000    0.000000
25%      0.000000   10.000000   10.000000   10.000000
50%      2.000000   11.000000   11.000000   12.000000
75%      6.000000   13.000000   13.000000   14.000000
max     32.000000   19.000000   19.000000   19.000000

Dataframe has no null values
columns: 33, rows: 649
```

Observations - There are 33 columns, including the target variable G3. - There are 649 observed students - Numeric columns show expected ranges and summary statistics. - There are no missing values, making preprocessing simpler.

0.3 Preprocessing the Data

0.3.1 Binary Encoding

Binary categorical features (e.g., sex, schoolsup) are mapped to 0/1 for modeling:

[36]:
```python
# ---------------  Convert Binary Nominal Features to Binary Numeric -----------------

# Define binary mappings
binary_mappings = {
    'school': {'GP': 1, 'MS': 0},
    'sex': {'M': 1, 'F': 0},
    'address': {'U': 1, 'R': 0},
    'famsize': {'GT3': 1, 'LE3': 0},
    'Pstatus': {'T': 1, 'A': 0},
    'schoolsup': {'yes': 1, 'no': 0},
    'famsup': {'yes': 1, 'no': 0},
    'paid': {'yes': 1, 'no': 0},
    'activities': {'yes': 1, 'no': 0},
    'nursery': {'yes': 1, 'no': 0},
    'higher': {'yes': 1, 'no': 0},
    'internet': {'yes': 1, 'no': 0},
    'romantic': {'yes': 1, 'no': 0}
}

# Make a copy of the raw dataframe
df = df_raw.copy()

# Apply mappings column by column and ensure integer type
for col, mapping in binary_mappings.items():
    if col in df.columns:
        df[col] = df[col].map(mapping).astype(int)

# Save the preprocessed dataset
df.to_csv('../data/student_preprocessed.csv', index=False)

# Display the updated dataframe
display(df.head())
display(df.info())
```

```
   school  sex  age  address  famsize  Pstatus  Medu  Fedu     Mjob      Fjob  \
0       1    0   18        1        1        0     4     4  at_home   teacher
1       1    0   17        1        1        1     1     1  at_home     other
2       1    0   15        1        0        1     1     1  at_home     other
3       1    0   15        1        1        1     4     2   health  services
4       1    0   16        1        1        1     3     3    other     other

   reason guardian  traveltime  studytime  failures  schoolsup  famsup  paid  \
0  course   mother           2          2         0          1       0     0
1  course   father           1          2         0          0       1     0
2   other   mother           1          2         0          1       0     0
3    home   mother           1          3         0          0       1     0
4    home   father           1          2         0          0       1     0

   activities  nursery  higher  internet  romantic  famrel  freetime  goout  \
0           0        1       1         0         0       4         3      4
1           0        0       1         1         0       5         3      3
2           0        1       1         1         0       4         3      2
3           1        1       1         1         1       3         2      2
4           0        1       1         0         0       4         3      2

   Dalc  Walc  health  absences  G1  G2  G3
0     1     1       3         4   0  11  11
1     1     1       3         2   9  11  11
```

```
2     2     3       3         6  12  13  12
3     1     1       5         0  14  14  14
4     1     2       5         0  11  13  13
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 649 entries, 0 to 648
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      649 non-null    int64
 1   sex         649 non-null    int64
 2   age         649 non-null    int64
 3   address     649 non-null    int64
 4   famsize     649 non-null    int64
 5   Pstatus     649 non-null    int64
 6   Medu        649 non-null    int64
 7   Fedu        649 non-null    int64
 8   Mjob        649 non-null    object
 9   Fjob        649 non-null    object
 10  reason      649 non-null    object
 11  guardian    649 non-null    object
 12  traveltime  649 non-null    int64
 13  studytime   649 non-null    int64
 14  failures    649 non-null    int64
 15  schoolsup   649 non-null    int64
 16  famsup      649 non-null    int64
 17  paid        649 non-null    int64
 18  activities  649 non-null    int64
 19  nursery     649 non-null    int64
 20  higher      649 non-null    int64
 21  internet    649 non-null    int64
 22  romantic    649 non-null    int64
 23  famrel      649 non-null    int64
 24  freetime    649 non-null    int64
 25  goout       649 non-null    int64
 26  Dalc        649 non-null    int64
 27  Walc        649 non-null    int64
 28  health      649 non-null    int64
 29  absences    649 non-null    int64
 30  G1          649 non-null    int64
 31  G2          649 non-null    int64
 32  G3          649 non-null    int64
dtypes: int64(29), object(4)
memory usage: 167.4+ KB
```

```
None
```

Explanation - Binary variables are converted for computation in models like OLS and tree-based algorithms. - Four multi-category nominal variables remain (dtype: object) and will be one-hot encoded later - Data set is clean and ready for analysis

0.4 Exploratory Data Analysis

[37]:
```
# Load the saved preprocessed dataset
df = pd.read_csv('../data/student_preprocessed.csv')

# --- Define variable types ---
```

```
# Binary Categorical variables (0/1)
bin_vars = ['school', 'sex', 'address', 'famsize', 'Pstatus', 'schoolsup', 'famsup', 'paid',␣
↪'activities', 'nursery',
'higher', 'internet', 'romantic']
# Ordinal Categorical variables (1-5 scale)
ord_vars = ['Medu', 'Fedu', 'traveltime', 'studytime', 'failures', 'famrel', 'freetime',␣
↪'goout', 'Dalc', 'Walc', 'health']
# Nominal Categorical variables (more than 2 categories)
nom_vars = ['Mjob', 'Fjob', 'reason', 'guardian']
# Continuous Numeric variables
cont_vars = ['age', 'absences', 'G1', 'G2', 'G3']
# Categorical variables (including binary and ordinal)
cat_vars = bin_vars + ord_vars + nom_vars
# Numeric variables (including continuous and ordinal)
numeric_vars = cont_vars + ord_vars
# All variables
all_vars = cat_vars + cont_vars
```

0.4.1 Distributions of Categorical Variables

We visualize the count of students for each category:

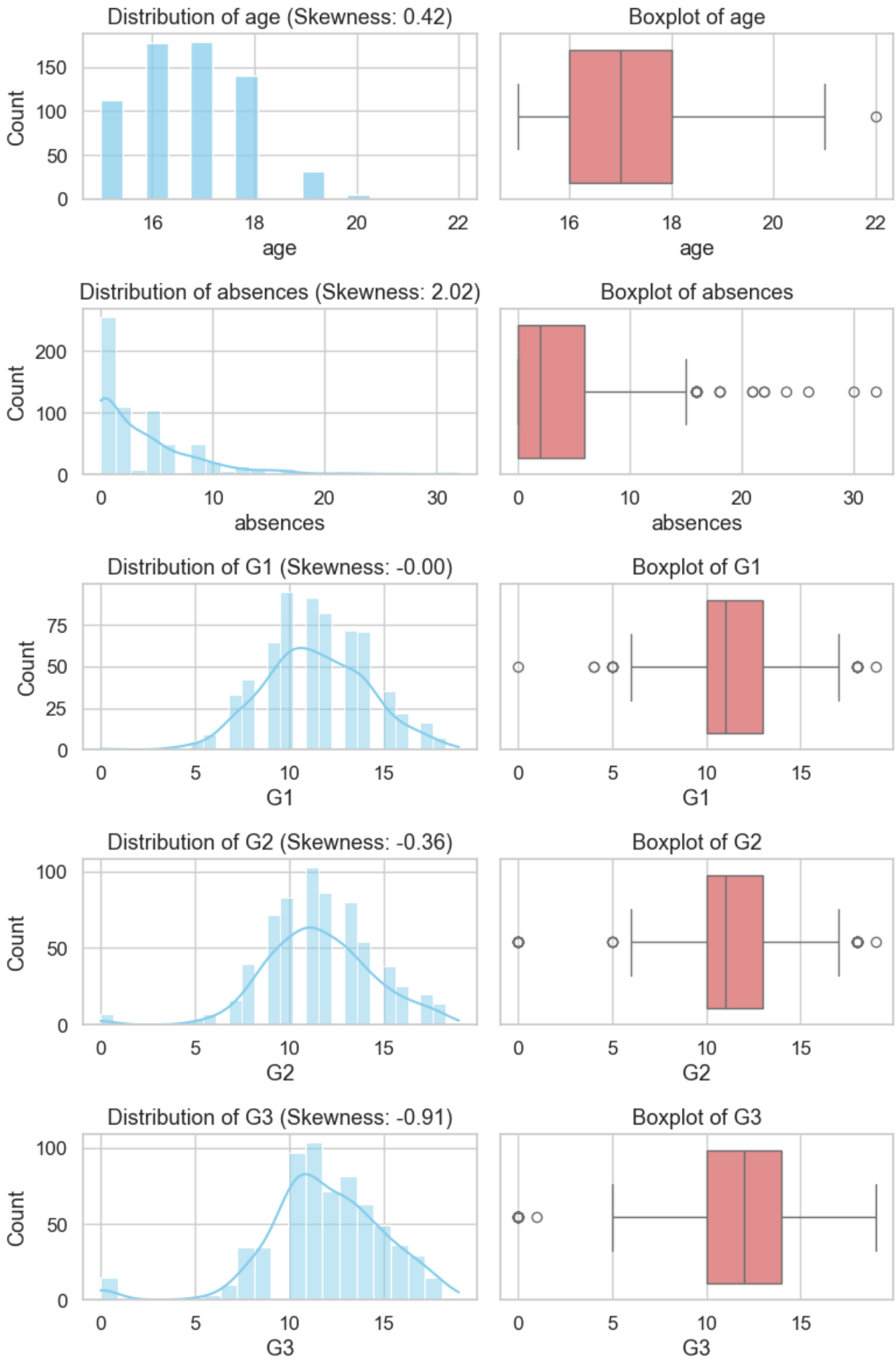
[38]:
```
# ------------- Count Plots for Categorical Variables -------------------

# Function to create chart titles
def make_title(var):
    return f"Counts by {var.replace('_', ' ').title()}"

# Plot setup
n_vars = len(cat_vars)
n_cols = 4
n_rows = int(np.ceil(n_vars / n_cols))
fig, axes = plt.subplots(n_rows, n_cols, figsize=(n_cols*4, n_rows*4))
axes = axes.flatten()

# Loop through categorical variables
for i, col in enumerate(cat_vars):
    sns.countplot(x=col, data=df_raw, hue=col, palette="pastel", ax=axes[i])        # Use␣
↪df_raw to show original categories
    axes[i].set_title(make_title(col))
    if axes[i].get_legend() is not None:
        axes[i].get_legend().remove()

# Remove unused subplots
for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
fig.savefig("../visualizations/categorical_counts.png", dpi=300, bbox_inches='tight')
plt.show()
```

Counts by School
count
GP
MS
school
Counts by Sex
count
F
M
sex
Counts by Address
count
U
R
address
Counts by Famsize
count
GT3
LE3
famsize
Counts by Pstatus
count
A
T
Pstatus
Counts by Schoolsup
count
yes
no
schoolsup
Counts by Famsup
count
no
yes
famsup
Counts by Paid
count
no
yes
paid
Counts by Activities
count
no
yes
activities
Counts by Nursery
count
yes
no
nursery
Counts by Higher
count
yes
no
higher
Counts by Internet
count
no
yes
internet
Counts by Romantic
count
no
yes
romantic
Counts by Medu
count
0
1
2
3
4
Medu
Counts by Fedu
count
0
1
2
3
4
Fedu
Counts by Traveltime
count
1
2
3
4
traveltime
Counts by Studytime
count
1
2
3
4
studytime
Counts by Failures
count
0
1
2
3
failures
Counts by Famrel
count
1
2
3
4
5
famrel
Counts by Freetime
count
1
2
3
4
5
freetime
Counts by Goout
count
1
2
3
4
5
goout
Counts by Dalc
count
1
2
3
4
5
Dalc
Counts by Walc
count
1
2
3
4
5
Walc
Counts by Health
count
1
2
3
4
5
health
Counts by Mjob
count
at_home
health
other
services
teacher
Mjob
Counts by Fjob
count
teacher
other
services
health
at_home
Fjob
Counts by Reason
count
course
other
home
reputation
reason
Counts by Guardian
count
mother
father
other
guardian

0.4.2 Distributions of Continuous Variables

We check distributions and skewness for continuous variables:

[39]:
```
# --------------- Distributions & Skewness of Continuous Variables ----------------

# Set style
sns.set_theme(style="whitegrid", palette="muted", font_scale=1.1)

# Plot distributions and boxplots for continuous variables
fig, axes = plt.subplots(len(cont_vars), 2, figsize=(8, 12))
for i, var in enumerate(cont_vars):
    kde_flag = False if var == 'age' else True  # No KDE for age due to discrete nature

    # Histogram with KDE
    sns.histplot(df[var], kde=kde_flag, ax=axes[i, 0], color="skyblue")
    skewness = df[var].skew()
    axes[i, 0].set_title(f"Distribution of {var} (Skewness: {skewness:.2f})")

    # Boxplot
    sns.boxplot(x=df[var], ax=axes[i, 1], color="lightcoral")
    axes[i, 1].set_title(f"Boxplot of {var}")

plt.tight_layout()
fig.savefig("../visualizations/continuous_distributions.png", dpi=300, bbox_inches='tight')
plt.show()
```


Distribution of age (Skewness: 0.42)
Count
age
Boxplot of age
age
Distribution of absences (Skewness: 2.02)
Count
absences
Boxplot of absences
absences
Distribution of G1 (Skewness: -0.00)
Count
G1
Boxplot of G1
G1
Distribution of G2 (Skewness: -0.36)
Count
G2
Boxplot of G2
G2
Distribution of G3 (Skewness: -0.91)
Count
G3
Boxplot of G3
G3

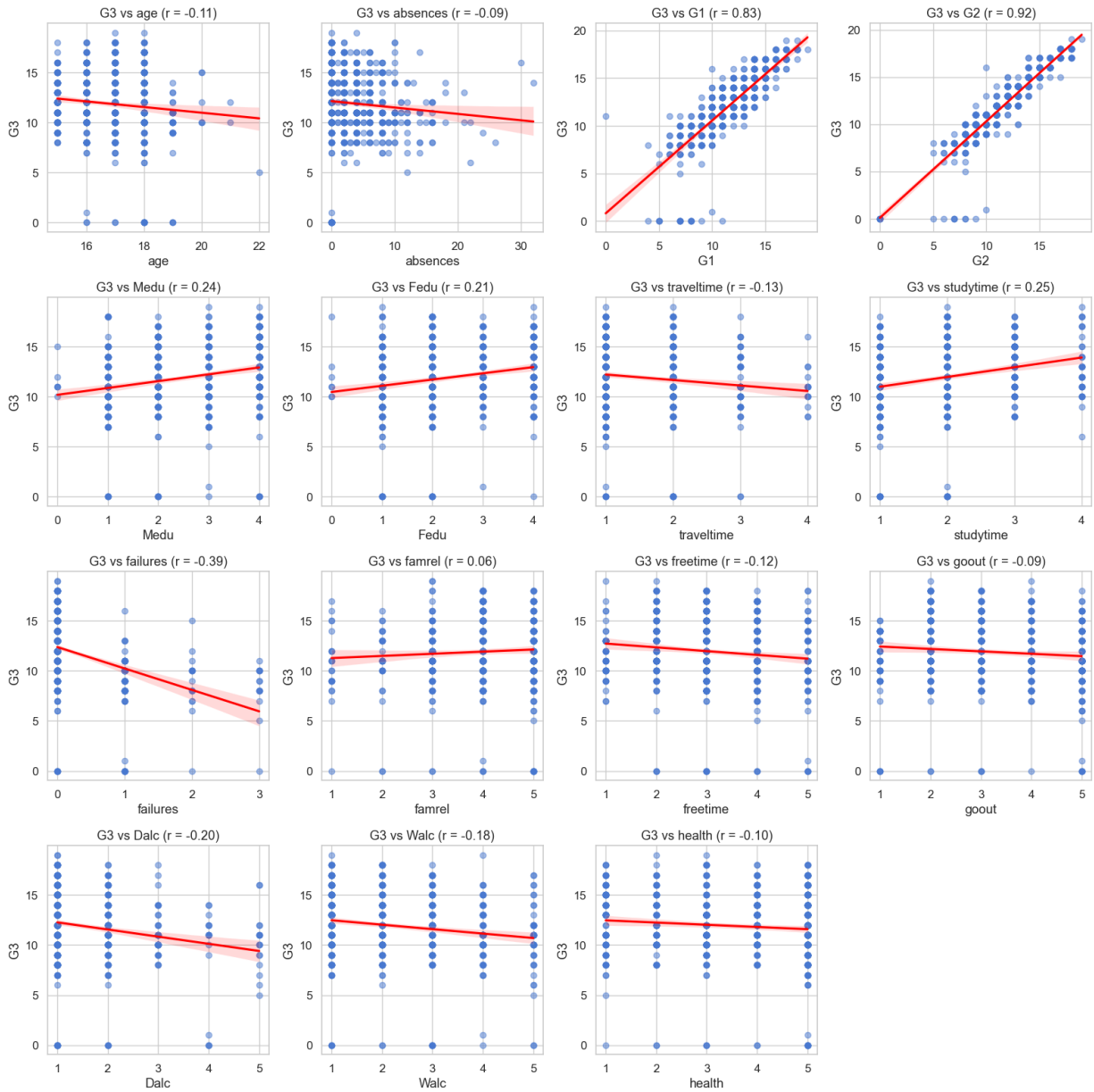
Observations - Most variables are approximately symmetric, except absences (right-skewed). - Boxplots reveal outliers in absences and some grades (G1, G2). - The target G3 is left-skewed, we will address this later

Interpretation - Skewed variables may require transformations in some models. - Outliers can affect OLS estimates; we'll consider removing them.

0.4.3 Numeric Relationships with G3

Scatterplots with regression lines show linear associations:

[40]:
```
# -------------- Scatter Plots & Regression vs G3 for Numeric Variables -----------------

# Remove 'G3' to avoid plotting vs itself
numeric_vars = [var for var in numeric_vars if var != 'G3']

# Determine grid size
n_vars = len(numeric_vars)
n_cols = 4
n_rows = int(np.ceil(n_vars / n_cols))

fig, axes = plt.subplots(n_rows, n_cols, figsize=(n_cols*4, n_rows*4))
axes = axes.flatten()

# Loop through variables and plot scatter + regression vs G3
for i, col in enumerate(numeric_vars):
    # Calculate correlation coefficient
    r = df[col].corr(df['G3'])

    sns.regplot(
        x=col,
        y='G3',
        data=df,
        scatter_kws={'alpha':0.5},
        line_kws={'color':'red'},
        ax=axes[i]
    )
    axes[i].set_title(f"G3 vs {col} (r = {r:.2f})")

# Remove any unused axes
for j in range(i+1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
fig.savefig("../visualizations/scatter_regression_g3.png", dpi=300, bbox_inches='tight')
plt.show()
```



Observations:

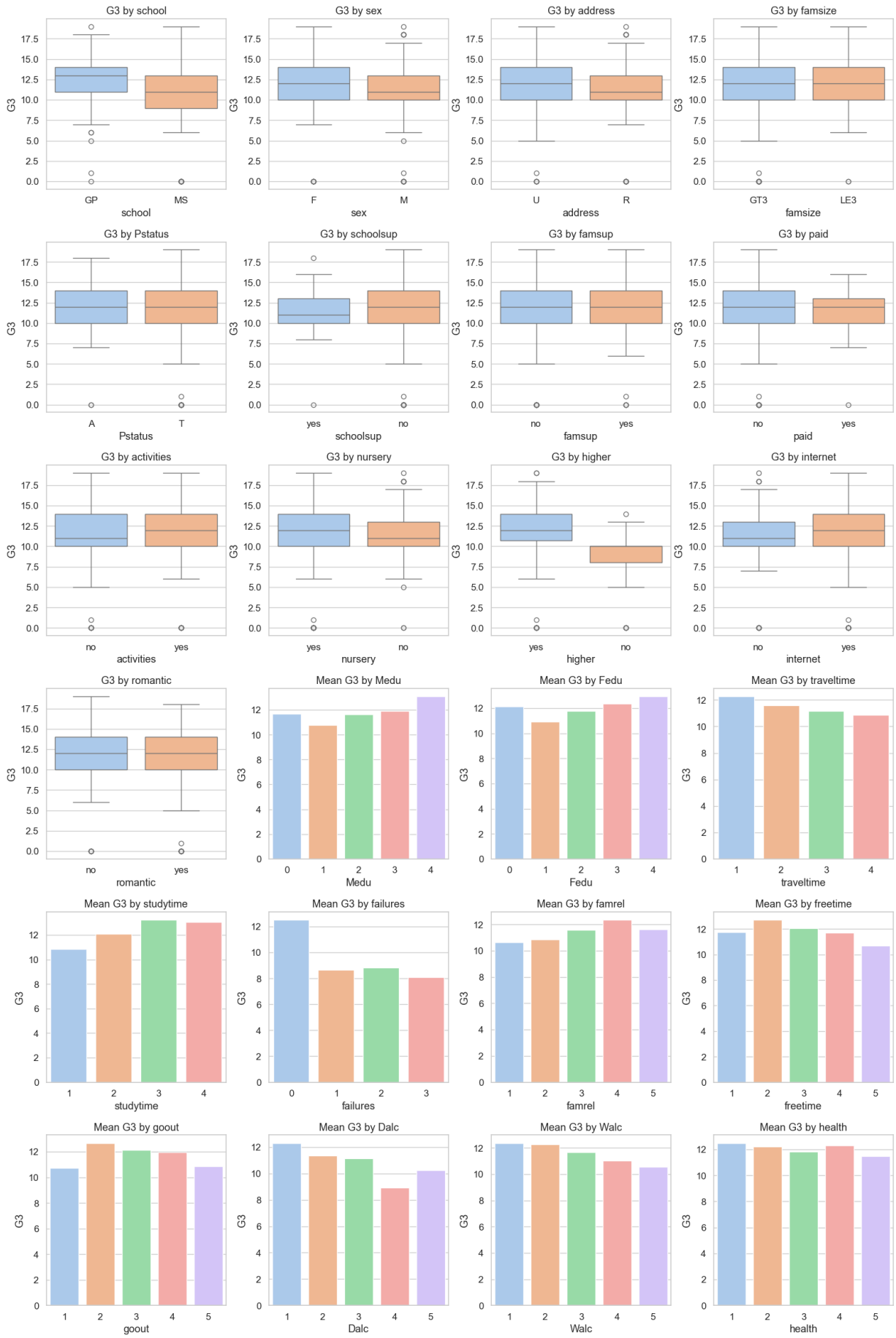
- G1 and G2 show the strongest correlation with G3 ($r \approx 0.83\&0.92$).
- Other variables like famrel show weak correlations.

0.4.4 Pairwise Comparision of Categorical Variables vs G3

Boxplots and barplots summarize mean performance by category:

[41]:
```
# -------------- Group Comparisons (Boxplots & Bar charts) ---------------
n_vars = len(bin_vars + ord_vars)
n_cols = 4
n_rows = int(np.ceil(n_vars / n_cols))

fig, axes = plt.subplots(n_rows, n_cols, figsize=(n_cols*4, n_rows*4))
axes = axes.flatten()
```

```python
for i, var in enumerate(bin_vars + ord_vars):
    ax = axes[i]

    # Binary variables (boxplot)
    if var in bin_vars:
        sns.boxplot(x=var, y='G3', data=df_raw, ax=ax, palette='pastel', hue=var)
        ax.set_title(f"G3 by {var}")

    # Ordinal variables (barplot of mean G3)
    elif var in ord_vars:
        sns.barplot(x=var, y='G3', data=df_raw, errorbar=None, ax=ax, palette='pastel', hue=var)
        ax.set_title(f"Mean G3 by {var}")

    # Remove legend
    legend = ax.get_legend()
    if legend is not None:
        legend.remove()

# Remove unused axes
for j in range(i+1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
fig.savefig("../visualizations/group_comparisons.png", dpi=300, bbox_inches='tight')
plt.show()
```

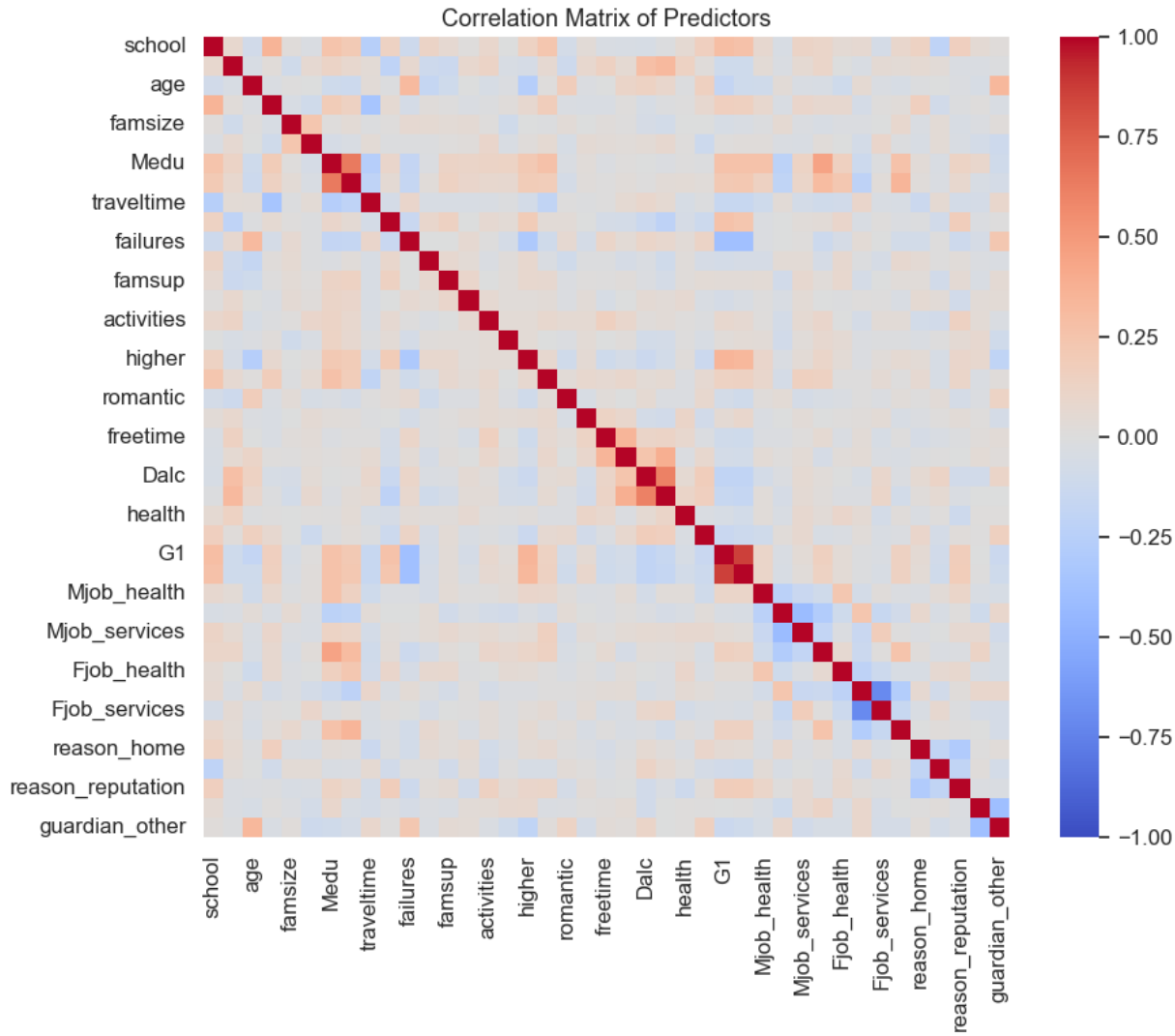

G3 by school
G3 by sex
G3 by address
G3 by famsize
G3 by Pstatus
G3 by schoolsup
G3 by famsup
G3 by paid
G3 by activities
G3 by nursery
G3 by higher
G3 by internet
G3 by romantic
Mean G3 by Medu
Mean G3 by Fedu
Mean G3 by traveltime
Mean G3 by studytime
Mean G3 by failures
Mean G3 by famrel
Mean G3 by freetime
Mean G3 by goout
Mean G3 by Dalc
Mean G3 by Walc
Mean G3 by health
G3
school
sex
address
famsize
Pstatus
schoolsup
famsup
paid
activities
nursery
higher
internet
romantic
Medu
Fedu
traveltime
studytime
failures
famrel
freetime
goout
Dalc
Walc
health

0.5 Correlation Analysis

0.5.1 Correlation Matrix

We examine correlations among numeric predictors:

[42]:
```
# ---------------- Correlation Matrix for Predictors ----------------

# Encode nominal categorical variables using one-hot encoding
df_encoded = df.copy()
df_encoded = pd.get_dummies(df_encoded, columns=nom_vars, drop_first=True)

# Convert any boolean columns to int
bool_cols = df_encoded.select_dtypes(include='bool').columns
df_encoded[bool_cols] = df_encoded[bool_cols].astype(int)

# Drop target variable to get predictors only
X = df_encoded.drop(columns='G3')  # predictors only

plt.figure(figsize=(10, 8))
sns.heatmap(
    X.corr(),
    cmap="coolwarm",
    vmin=-1, vmax=1,
    annot=False
)
plt.title("Correlation Matrix of Predictors")
plt.savefig("../visualizations/correlation_matrix.png", dpi=300, bbox_inches='tight')
plt.show()
```



Observations

- High correlation between G1 and G2 may indicate multicollinearity.
- Helps guide feature selection and VIF checks later.

0.5.2 Association Strength for Categorical Variables

We compute Pearson correlations for binary and eta-squared for multi-category features:

[43]:
```
# ---------------- Association Strength for Categorical Variables ----------------

# Identify categorical variables
categorical_cols = df[cat_vars]

results = []

for col in cat_vars:
    unique_vals = df[col].dropna().unique()

    # Binary variable → use Pearson (point-biserial)
    if len(unique_vals) == 2 and set(unique_vals).issubset({0, 1}):
```

```
        corr, pval = pearsonr(df[col], df['G3'])
        results.append({
            'Variable': col,
            'Type': 'Binary',
            'Correlation': corr,
            'p-value': pval
        })

    # Multi-category variable → use eta-squared (from ANOVA)
    elif len(unique_vals) > 2:
        groups = [df.loc[df[col] == val, 'G3'].dropna() for val in unique_vals]
        f_stat, pval = f_oneway(*groups)
        ss_between = sum(len(g) * (g.mean() - df['G3'].mean())**2 for g in groups)
        ss_total = sum((df['G3'] - df['G3'].mean())**2)
        eta_sq = ss_between / ss_total if ss_total != 0 else np.nan
        results.append({'Variable': col, 'Type': 'Multi-Category', 'Correlation': eta_sq,␣
↪'p-value': pval})

# Convert to DataFrame and sort
results_df = pd.DataFrame(results)
results_df['|Correlation|'] = results_df['Correlation'].abs()
results_df = results_df.sort_values(by='|Correlation|', ascending=False)

print("Strength of association between categorical variables and G3:")
print(results_df[['Variable', 'Type', 'Correlation', 'p-value']])
```

```
Strength of association between categorical variables and G3:
      Variable            Type  Correlation       p-value
10      higher          Binary     0.332172  3.499660e-18
0       school          Binary     0.284294  1.566199e-13
17    failures  Multi-Category     0.192900  8.644516e-30
2      address          Binary     0.167637  1.764153e-05
11    internet          Binary     0.150025  1.248917e-04
1          sex          Binary    -0.129077  9.815287e-04
12    romantic          Binary    -0.090583  2.100247e-02
16   studytime  Multi-Category     0.068765  5.705728e-10
5    schoolsup          Binary    -0.066405  9.097104e-02
13        Medu  Multi-Category     0.062669  1.885685e-08
8   activities          Binary     0.059791  1.280991e-01
6       famsup          Binary     0.059206  1.318865e-01
7         paid          Binary    -0.054898  1.624412e-01
14        Fedu  Multi-Category     0.048595  1.799114e-06
21        Dalc  Multi-Category     0.048425  1.898967e-06
26      reason  Multi-Category     0.045498  1.341642e-06
3      famsize          Binary    -0.045016  2.521332e-01
24        Mjob  Multi-Category     0.043774  8.305150e-06
20       goout  Multi-Category     0.041586  1.652480e-05
22        Walc  Multi-Category     0.033239  2.193385e-04
9      nursery          Binary     0.028752  4.646452e-01
19    freetime  Multi-Category     0.027123  1.389612e-03
18      famrel  Multi-Category     0.022334  5.684831e-03
25        Fjob  Multi-Category     0.019922  1.137628e-02
15  traveltime  Multi-Category     0.016733  1.232110e-02
23      health  Multi-Category     0.014931  4.574054e-02
27    guardian  Multi-Category     0.008102  7.226240e-02
4      Pstatus          Binary    -0.000754  9.847058e-01
```

Observations

We see strong correlation between G3 and higher (pursuit of higher education), school, failures, and address (rural or urban)

0.5.3 Association of Strength for Numeric Variables

[44]:
```
# ----------------- Strength of Association for Numeric (Continuos) Variables ----------------

#  Numeric variables: Pearson correlation with G3
# Dictionary to store results
numeric_results = {}

for col in cont_vars:  # list of numeric predictors
    corr, pval = pearsonr(df[col], df['G3'])
    numeric_results[col] = {'Correlation': corr, 'p-value': pval}

# Convert to DataFrame
numeric_corr_df = pd.DataFrame(numeric_results).T  # transpose so variables are rows
numeric_corr_df = numeric_corr_df.sort_values(by='Correlation', key=abs, ascending=False)  #␣
↪sort by absolute correlation

print(numeric_corr_df)
```

```
          Correlation        p-value
G3           1.000000   0.000000e+00
G2           0.918548  5.642401e-263
G1           0.826387  1.617571e-163
age         -0.106505   6.612115e-03
absences    -0.091379   1.989562e-02
```

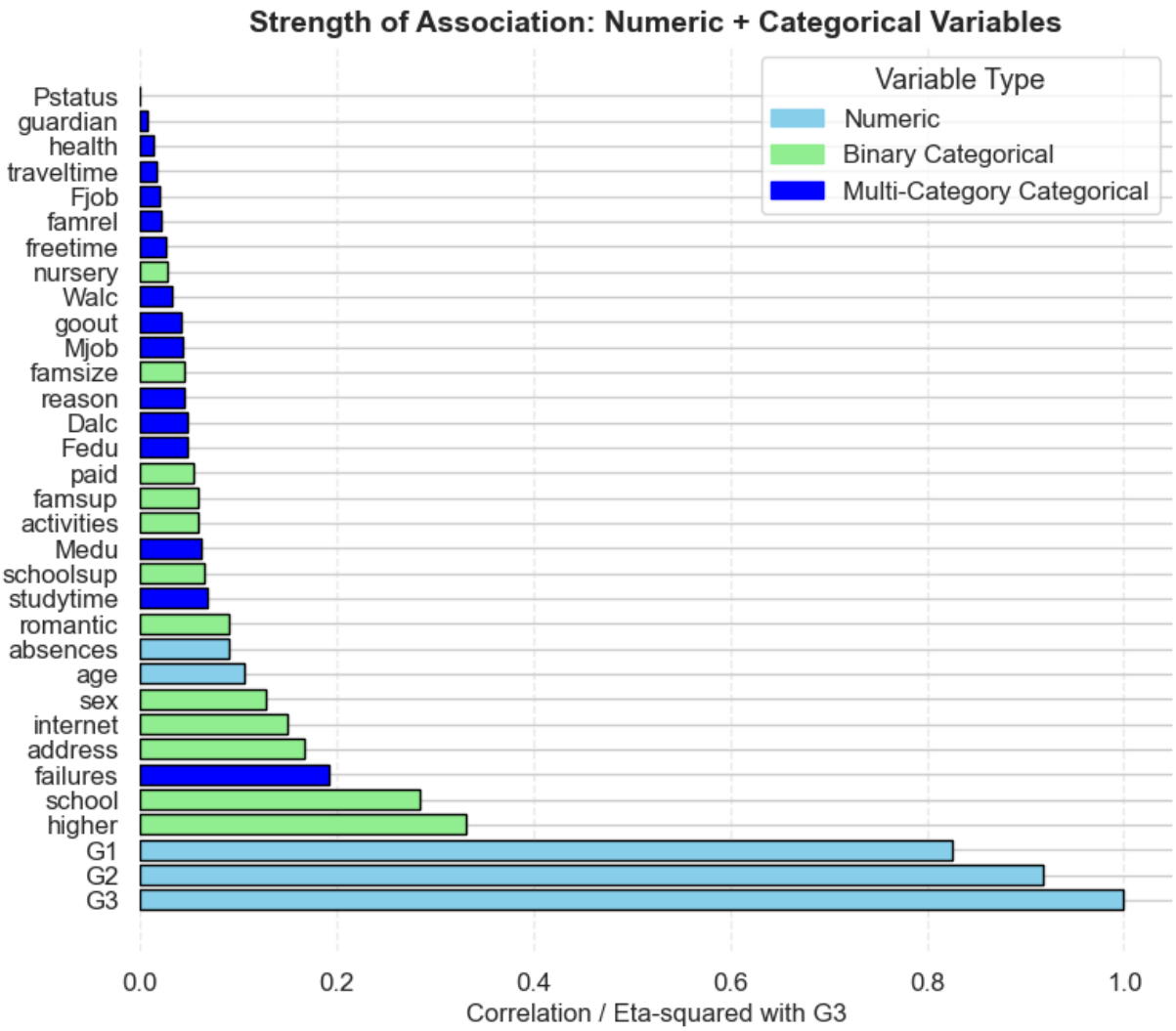
Observations

We see strong correlation between G3 and G1, G2

0.5.4 Combined Association of Strength Plot

[45]:
```
#  ----------------- Combined Bar Plot of Numeric + Categorical Associations ------------------

# Convert numeric_corr_df to match structure of results_df
numeric_plot_df = numeric_corr_df.copy()
numeric_plot_df['Variable'] = numeric_plot_df.index
numeric_plot_df['Type'] = 'Numeric'
numeric_plot_df = numeric_plot_df.reset_index(drop=True)

categorical_plot_df = results_df.copy()
categorical_plot_df = categorical_plot_df.rename(columns={'Correlation': 'Correlation',␣
↪'Variable': 'Variable'})

# Combine
combined_df = pd.concat([numeric_plot_df[['Variable', 'Type', 'Correlation']],
                         categorical_plot_df[['Variable', 'Type', 'Correlation']]])

# Sort for plotting
combined_df['Correlation'] = combined_df['Correlation'].abs()
combined_df = combined_df.sort_values(by='Correlation', key=abs, ascending=True)

# Colors
colors = combined_df['Type'].map({'Numeric': 'skyblue', 'Binary': 'lightgreen',␣
↪'Multi-Category': 'blue'})
```

```python
# Create legend patches
legend_patches = [
    mpatches.Patch(color='skyblue', label='Numeric'),
    mpatches.Patch(color='lightgreen', label='Binary Categorical'),
    mpatches.Patch(color='blue', label='Multi-Category Categorical')
]

# Plot
sns.set_theme(style="whitegrid", palette="muted", font_scale=1.1)

plt.figure(figsize=(8, 7))
plt.barh(
    combined_df['Variable'],
    combined_df['Correlation'],
    color=colors,
    edgecolor='black'
)
plt.xlabel('Correlation / Eta-squared with G3', fontsize=12)
plt.title('Strength of Association: Numeric + Categorical Variables', fontsize=14,␣
 ↪weight='bold')
plt.gca().invert_yaxis()
plt.grid(axis='x', linestyle='--', alpha=0.4)
plt.legend(handles=legend_patches, title='Variable Type', loc='upper right')
plt.tight_layout()
sns.despine(left=True, bottom=True)
plt.savefig("../visualizations/combined_association_strength.png", dpi=300, bbox_inches='tight')
plt.show()
```



Explanation:

- Shows the absolute correlation / eta-squared values across all variables.
- Facilitates quick identification of the most influential features.

0.5.5 Chi Square Tests for Categorical Variables

[46]:
```
# --------------- Chi-square Tests for Categorical Variables ----------------

alpha = 0.05  # significance threshold

chi2_results = []

for i in range(len(cat_vars)):
    for j in range(i+1, len(cat_vars)):
        var1 = cat_vars[i]
        var2 = cat_vars[j]

        # Contingency table
        table = pd.crosstab(df[var1], df[var2])
        chi2, p_val, dof, expected = chi2_contingency(table)
```

```
        # Cramer's V effect size
        n = table.sum().sum()
        min_dim = min(table.shape) - 1
        cramers_v = np.sqrt(chi2 / (n * min_dim)) if min_dim > 0 else np.nan

        chi2_results.append({
            "Variable 1": var1,
            "Variable 2": var2,
            "Chi2": round(chi2, 3),
            "p-value": round(p_val, 5),
            "dof": dof,
            "Cramer's V": round(cramers_v, 3)
        })

chi2_df = pd.DataFrame(chi2_results)
chi2_df = chi2_df.sort_values("p-value").reset_index(drop=True)

print(f"Significant Chi-square results (p < {alpha}):")
display(chi2_df[chi2_df["p-value"] < alpha])
print("Largest effect sizes (Cramer's V):")
display(chi2_df.sort_values("Cramer's V", ascending=False).head(10))
```

```
Significant Chi-square results (p < 0.05):

     Variable 1 Variable 2     Chi2  p-value  dof  Cramer's V
0          Dalc       Walc  442.393  0.00000   16       0.413
1        higher  studytime   34.300  0.00000    3       0.230
2        school       Mjob   37.876  0.00000    4       0.242
3        school     reason   52.524  0.00000    3       0.284
4        higher       Medu   30.594  0.00000    4       0.217
..          ...        ...      ...      ...  ...         ...
115      school        sex    4.129  0.04216    1       0.080
116        Medu   guardian   15.925  0.04347    8       0.111
117  activities   internet    4.022  0.04490    1       0.079
118    romantic       Dalc    9.642  0.04692    4       0.122
119   schoolsup     higher    3.867  0.04923    1       0.077

[120 rows x 6 columns]

Largest effect sizes (Cramer's V):

   Variable 1  Variable 2     Chi2  p-value  dof  Cramer's V
0        Dalc        Walc  442.393      0.0   16       0.413
6        Medu        Fedu  388.485      0.0   16       0.387
13       Medu        Mjob  378.368      0.0   16       0.382
16    address  traveltime   81.214      0.0    3       0.354
22     school     address   79.959      0.0    1       0.351
12        sex        Walc   75.113      0.0    4       0.340
18     higher    failures   67.074      0.0    3       0.321
30     school  traveltime   61.120      0.0    3       0.307
8    internet        Mjob   59.580      0.0    4       0.303
11        sex        Dalc   57.774      0.0    4       0.298
```

Interpretation

Althought there are several significant tests to say there may be some correlation between variables, the highest effect size (Cramer's V) is 0.413, which is only a moderate strength of association. We shouldn't have to worry about strong associastions here.

0.6 Inferential Statistics

0.6.1 T-tests for Binary Categorical Variables vs G3

[47]:
```
# ---------------- T-tests for Binary Categorical vs G3 ----------------

alpha = 0.05  # significance threshold

t_results = []

target = 'G3'  # target variable

for var in bin_vars:  # binary predictors
    group0 = df[df[var] == 0][target]
    group1 = df[df[var] == 1][target]

    # Variance ratio
    var0 = np.var(group0, ddof=1)
    var1 = np.var(group1, ddof=1)
    ratio = max(var0, var1) / min(var0, var1)

    # Choose t-test type
    if ratio > 1.5:
        t_stat, p_val = ttest_ind(group0, group1, equal_var=False)
        test_type = "Welch's t-test"
    else:
        t_stat, p_val = ttest_ind(group0, group1, equal_var=True)
        test_type = "Student's t-test"

    # Effect size (Cohen's d)
    mean_diff = group1.mean() - group0.mean()
    pooled_sd = np.sqrt((var0 + var1) / 2)
    cohens_d = mean_diff / pooled_sd

    # Append results
    t_results.append({
        "Binary Variable": var,
        "Target Variable": target,
        "Test Type": test_type,
        "t-statistic": round(t_stat, 3),
        "p-value": round(p_val, 5),
        "Variance Ratio": round(ratio, 2),
        "Group0 Mean": round(group0.mean(), 2),
        "Group1 Mean": round(group1.mean(), 2),
        "Cohen's d": round(cohens_d, 3)
    })

# Convert to DataFrame
t_results_df = pd.DataFrame(t_results)
t_results_df = t_results_df.sort_values("p-value").reset_index(drop=True)

print(f"Significant t-test results (p < {alpha}):")
display(t_results_df[t_results_df["p-value"] < alpha])
```

```
Significant t-test results (p < 0.05):

  Binary Variable Target Variable         Test Type  t-statistic  p-value  \
0          school              G3    Welch's t-test       -6.754  0.00000
1          higher              G3  Student's t-test       -8.958  0.00000
```

```
2         address              G3  Student's t-test       -4.325  0.00002
3        internet              G3  Student's t-test       -3.860  0.00012
4             sex              G3  Student's t-test        3.311  0.00098
5        romantic              G3  Student's t-test        2.314  0.02100
6       schoolsup              G3    Welch's t-test        2.247  0.02675

   Variance Ratio  Group0 Mean  Group1 Mean  Cohen's d
0            2.13        10.65        12.58      0.586
1            1.06         8.80        12.28      1.153
2            1.46        11.09        12.26      0.355
3            1.22        11.03        12.17      0.349
4            1.13        12.25        11.41     -0.263
5            1.41        12.13        11.52     -0.184
6            2.07        11.98        11.28     -0.245
```

Observations

Students who attend GP school, have internet access, plan for higher education, and live in urban areas tend to achieve significantly higher final grades (G3). Conversely, those receiving school support, in romantic relationships, or female students show slightly lower average grades, though these effects are smaller in magnitude.

0.6.2 ANOVA Tests for Multi-level Categorical Variables vs G3

[48]:
```python
# -------------- ANOVA for Multi-level Categorical vs Continuous ----------------

alpha = 0.05
target = 'G3'
anova_results = []

for var in ord_vars:  # ordinal/multi-level categorical predictors
    # Skip variables with less than 2 levels
    if df[var].nunique() < 2:
        continue

    # Fit ANOVA model
    model = ols(f"{target} ~ C({var})", data=df).fit()
    anova_table = sm.stats.anova_lm(model, typ=2)

    f_stat = anova_table.loc[f"C({var})", "F"]
    p_val = anova_table.loc[f"C({var})", "PR(>F)"]

    # Eta-squared: sum_sq_var / sum_sq_total
    eta_sq = anova_table.loc[f"C({var})", "sum_sq"] / anova_table["sum_sq"].sum()

    anova_results.append({
        "Categorical Variable": var,
        "Target Variable": target,
        "F-statistic": round(f_stat, 3),
        "p-value": round(p_val, 5),
        "Eta-squared": round(eta_sq, 3)
    })

anova_df = pd.DataFrame(anova_results).sort_values("p-value").reset_index(drop=True)

print(f"Significant ANOVA results (p < {alpha}):")
display(anova_df[anova_df["p-value"] < alpha])
```

```
Significant ANOVA results (p < 0.05):
```

```
   Categorical Variable Target Variable  F-statistic  p-value  Eta-squared
0                  Medu              G3       10.764  0.00000        0.063
1                  Fedu              G3        8.223  0.00000        0.049
2             studytime              G3       15.876  0.00000        0.069
3              failures              G3       51.386  0.00000        0.193
4                  Dalc              G3        8.193  0.00000        0.048
5                 goout              G3        6.986  0.00002        0.042
6                  Walc              G3        5.535  0.00022        0.033
7              freetime              G3        4.489  0.00139        0.027
8                famrel              G3        3.678  0.00568        0.022
9            traveltime              G3        3.659  0.01232        0.017
10               health              G3        2.440  0.04574        0.015
```

Observations

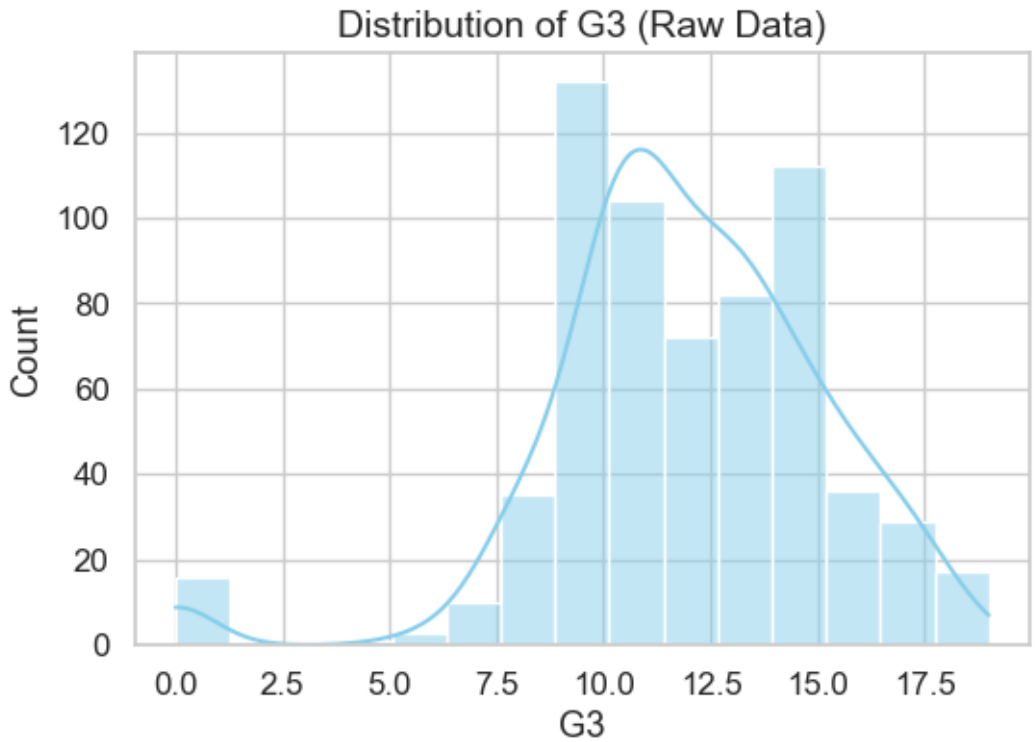
Students' final grades (G3) vary significantly across several multi-level factors: higher parental education, more study time, and fewer past failures are strongly linked to better performance, while frequent alcohol use, more going out, and longer travel times are associated with lower grades. Overall, failures and studytime show the largest effect sizes ($\eta \approx 0.19$ and 0.07).

0.7 Checking Model Assumptions

0.7.1 Normal Distribution of Target Variable

We being by checking if the target variable 'G3' is normally distributed:

[49]:
```
# ------------------ Check Distribution of 'G3' ------------------

target = 'G3'

# Plot distribution of target variable (raw)
plt.figure(figsize=(6,4))
sns.histplot(df[target], kde=True, color='skyblue', bins=15)
plt.title(f"Distribution of {target} (Raw Data)", fontsize=14)
plt.xlabel(target)
plt.ylabel("Count")
plt.savefig("../visualizations/g3_distribution.png", dpi=300, bbox_inches='tight')
plt.show()

# Skewness and kurtosis (raw)
skew_val_raw = df[target].skew()
kurt_val_raw = df[target].kurtosis()
print(f"Raw skewness: {skew_val_raw:.2f} | Raw kurtosis: {kurt_val_raw:.2f}")

# Shapiro-Wilk test (raw)
shapiro_stat, shapiro_p = stats.shapiro(df[target])
print(f"Shapiro-Wilk (raw): W = {shapiro_stat:.3f}, p = {shapiro_p:.4f}")
```




```
Raw skewness: -0.91 | Raw kurtosis: 2.71
Shapiro-Wilk (raw): W = 0.926, p = 0.0000
```

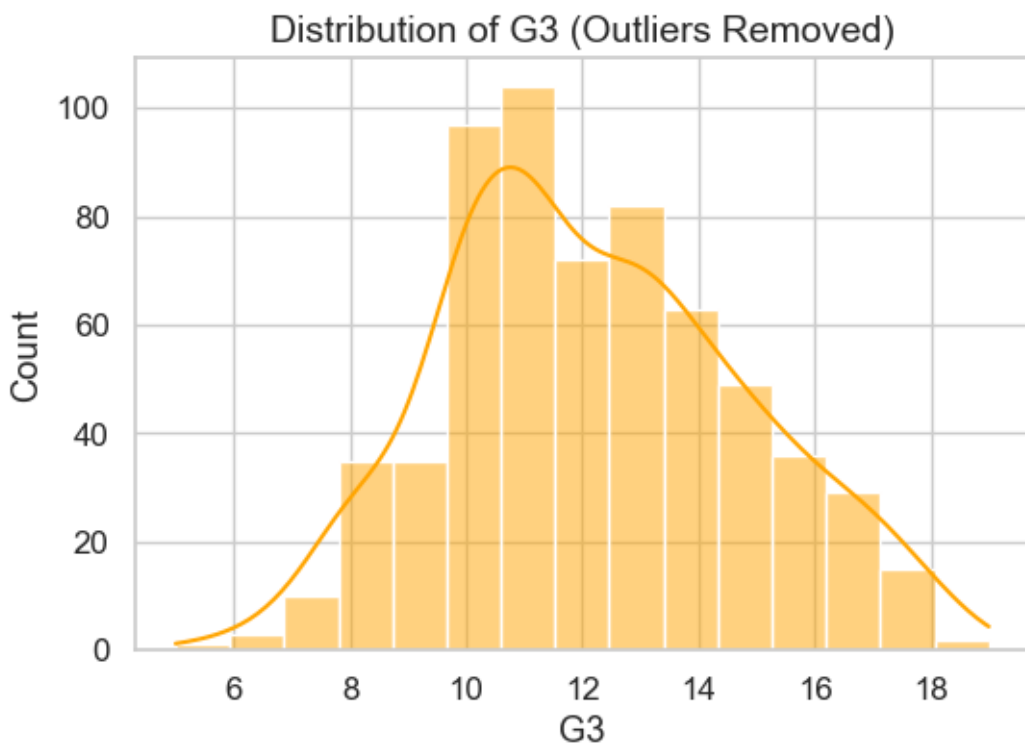
Interpretation

G3 is clearly right-skewed, we will need to see if removing the outliers makes the distribution more normal.

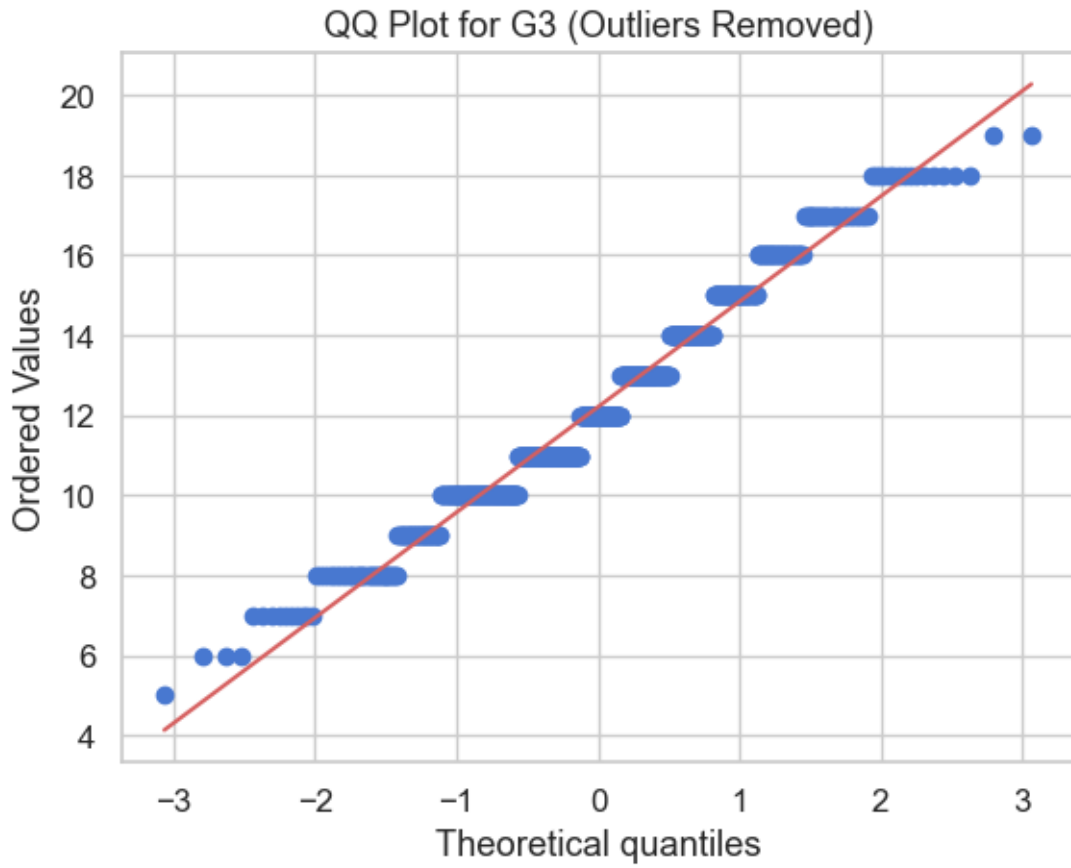
[50]:
```
# ------------------ Remove G3 Outliers and Check Distribution Again------------------

# Outlier Detection (IQR method)
Q1 = df[target].quantile(0.25)
Q3 = df[target].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df[target] < lower_bound) | (df[target] > upper_bound)]
print(f"\nNumber of potential outliers: {len(outliers)}")

# Remove outliers
df_no_outliers = df[(df[target] >= lower_bound) & (df[target] <= upper_bound)]

# Distribution and Normality (after outlier remova)
plt.figure(figsize=(6,4))
sns.histplot(df_no_outliers[target], kde=True, color='orange', bins=15)
plt.title(f"Distribution of {target} (Outliers Removed)", fontsize=14)
plt.savefig("../visualizations/g3_distribution_no_outliers.png", dpi=300, bbox_inches='tight')
plt.show()

# Skewness and kurtosis (after outlier removal)
skew_val = df_no_outliers[target].skew()
```

```
kurt_val = df_no_outliers[target].kurtosis()
print(f"After removing outliers: skewness = {skew_val:.2f}, kurtosis = {kurt_val:.2f}")

# Shapiro-Wilk (after outlier removal)
shapiro_stat, shapiro_p = stats.shapiro(df_no_outliers[target])
print(f"Shapiro-Wilk (no outliers): W = {shapiro_stat:.3f}, p = {shapiro_p:.4f}")

# QQ plot for visual check
stats.probplot(df_no_outliers[target], dist="norm", plot=plt)
plt.title(f"QQ Plot for {target} (Outliers Removed)")
plt.savefig("../visualizations/g3_qqplot_no_outliers.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
Number of potential outliers: 16
```




```
After removing outliers: skewness = 0.23, kurtosis = -0.47
Shapiro-Wilk (no outliers): W = 0.976, p = 0.0000
```



Observations

Removing the outliers helped shape G3 to be more normally distributed. G3 shifted from heavily left skewed to slightly right skewed. Even though G3 with outliers removed still doesn't pass the Shapiro-Wilk significance test, the QQ plot for G3 gives enough reason to suggest we can assume a normally distributed target variable

0.7.2 Multicollinearity of Categorical Predictors

[51]:
```
# ------------------ Check Multicollinearity (VIF) with One-hot Encoded Multi-category␣
↪Predictors ------------------

# X = predictors only (after one-hot encoding)
X = df_encoded.drop(columns=['G3'])
X = sm.add_constant(X)  # add intercept

# Compute VIF
vif_data = pd.DataFrame()
vif_data["Variable"] = X.columns
vif_data["VIF"] = [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]

# Sort descending for readability
vif_data = vif_data.sort_values(by="VIF", ascending=False).reset_index(drop=True)

# Display top 10 largest VIFs
display(vif_data.head(10))
```

```
        Variable         VIF
```

```
0          const  378.218504
1     Fjob_other    4.677847
2             G1    4.430493
3             G2    4.307080
4  Fjob_services    4.227396
5           Medu    2.693316
6   Mjob_teacher    2.277617
7   Fjob_teacher    2.178296
8           Walc    2.113194
9  Mjob_services    2.113154
```

Interpretation

The variance inflation factor (VIF) analysis indicates that multicollinearity among the predictor variables is generally low. All predictors exhibit VIF values below 5, suggesting that redundancy between variables is minimal and should not adversely affect the stability of the regression estimates. The exceptionally high VIF for the intercept (const) is expected and does not indicate multicollinearity among the predictors.

0.7.3 Residual Distribution of Numeric Variables

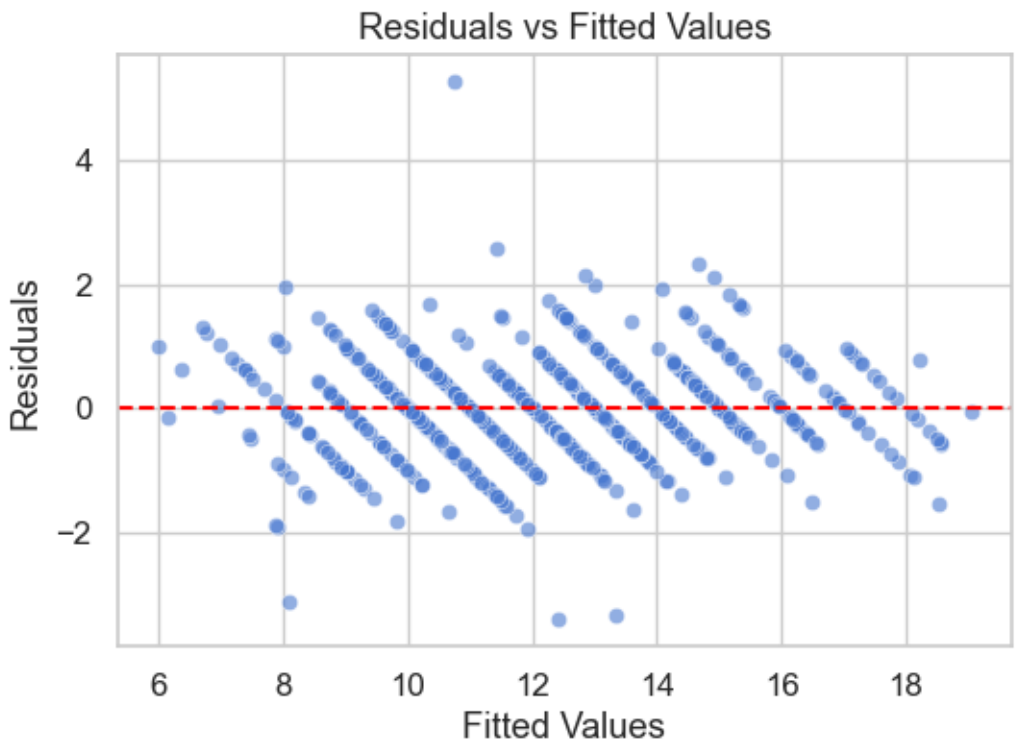
[60]:
```
# ------------------ Fit Preliminary OLS Model to Check Residuals ------------------

# Predictors (with one-hot encoding) and target
X = df_encoded.drop(columns=['G3'])
X = sm.add_constant(X)              # add intercept
y = df_no_outliers['G3']            # remove outliers

# Fit model
prelim_model = sm.OLS(y, X.loc[y.index]).fit()  # align indices since using outlier-cleaned y

# Residuals
residuals = prelim_model.resid
fitted = prelim_model.fittedvalues

# Summary
print(prelim_model.summary())

# Residual Plots

plt.figure(figsize=(6,4))
sns.scatterplot(x=fitted, y=residuals, alpha=0.6)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted Values")
plt.savefig("../visualizations/residuals_vs_fitted.png", dpi=300, bbox_inches='tight')
plt.show()

# QQ Plot of residuals
sm.qqplot(residuals, line='45', fit=True)
plt.title("QQ Plot of Residuals")
plt.savefig("../visualizations/qqplot_residuals.png", dpi=300, bbox_inches='tight')
plt.show()

# Histogram of residuals
plt.figure(figsize=(6,4))
sns.histplot(residuals, kde=True, color='purple', bins=15)
plt.xlabel("Residuals")
```

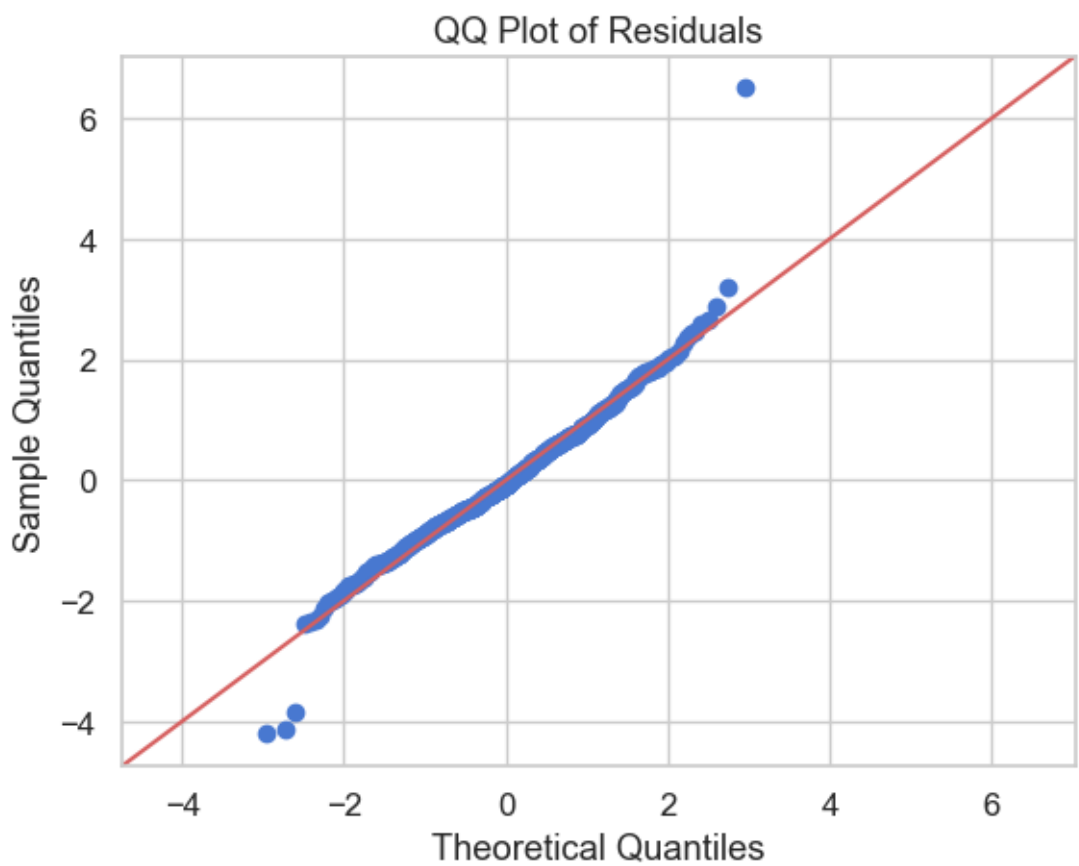
```
plt.ylabel("Count")
plt.title("Residual Distribution")
plt.savefig("../visualizations/residuals_distribution.png", dpi=300, bbox_inches='tight')
plt.show()
```

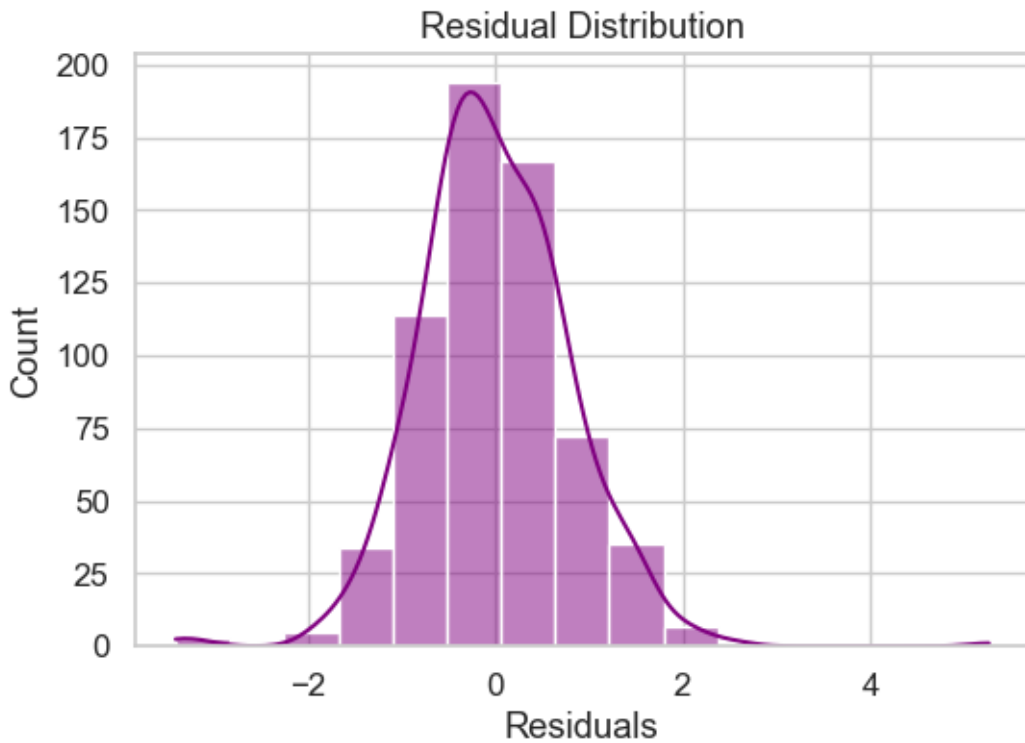
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                     G3   R-squared:                       0.907
Model:                            OLS   Adj. R-squared:                  0.901
Method:                 Least Squares   F-statistic:                     140.9
Date:                Thu, 16 Oct 2025   Prob (F-statistic):          6.51e-276
Time:                        14:42:04   Log-Likelihood:                -763.79
No. Observations:                 633   AIC:                             1612.
Df Residuals:                     591   BIC:                             1799.
Df Model:                          41
Covariance Type:            nonrobust
==============================================================================
=====
                        coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
-----
const                -0.3629      0.645     -0.563      0.574      -1.629
0.904
school                0.1379      0.087      1.590      0.112      -0.032
0.308
sex                  -0.1052      0.080     -1.313      0.190      -0.263
0.052
age                   0.1112      0.033      3.354      0.001       0.046
0.176
address               0.0082      0.084      0.098      0.922      -0.156
0.172
famsize               0.0625      0.078      0.804      0.422      -0.090
0.215
Pstatus              -0.1147      0.110     -1.039      0.299      -0.332
0.102
Medu                  0.0175      0.048      0.363      0.717      -0.077
0.112
Fedu                 -0.0141      0.044     -0.322      0.748      -0.100
0.072
traveltime            0.1224      0.050      2.426      0.016       0.023
0.221
studytime             0.0187      0.044      0.420      0.674      -0.069
0.106
failures             -0.1632      0.068     -2.403      0.017      -0.297
-0.030
schoolsup            -0.1917      0.117     -1.632      0.103      -0.422
0.039
famsup                0.0068      0.073      0.093      0.926      -0.136
0.149
paid                 -0.1830      0.147     -1.244      0.214      -0.472
0.106
activities           -0.0166      0.071     -0.233      0.816      -0.157
0.124
nursery              -0.0282      0.086     -0.326      0.744      -0.198
0.141
higher                0.2954      0.126      2.353      0.019       0.049
0.542
```

```
internet              0.1239      0.088      1.403      0.161      -0.050
0.297
romantic              0.0224      0.073      0.306      0.760      -0.121
0.166
famrel                0.0020      0.037      0.054      0.957      -0.072
0.076
freetime             -0.0137      0.036     -0.383      0.702      -0.084
0.056
goout                -0.0313      0.034     -0.910      0.363      -0.099
0.036
Dalc                 -0.0262      0.050     -0.526      0.599      -0.124
0.072
Walc                 -0.0013      0.038     -0.035      0.972      -0.076
0.073
health               -0.0363      0.025     -1.470      0.142      -0.085
0.012
absences             -0.0164      0.008     -2.046      0.041      -0.032
-0.001
G1                    0.1892      0.028      6.780      0.000       0.134
0.244
G2                    0.7374      0.028     25.937      0.000       0.682
0.793
Mjob_health           0.1488      0.169      0.878      0.380      -0.184
0.482
Mjob_other            0.1102      0.097      1.141      0.255      -0.080
0.300
Mjob_services         0.1026      0.118      0.868      0.386      -0.130
0.335
Mjob_teacher          0.2626      0.160      1.644      0.101      -0.051
0.576
Fjob_health          -0.3822      0.237     -1.613      0.107      -0.848
0.083
Fjob_other           -0.3478      0.145     -2.398      0.017      -0.633
-0.063
Fjob_services        -0.3571      0.153     -2.340      0.020      -0.657
-0.057
Fjob_teacher         -0.3151      0.214     -1.473      0.141      -0.735
0.105
reason_home           0.0396      0.091      0.436      0.663      -0.139
0.218
reason_other         -0.1323      0.119     -1.112      0.267      -0.366
0.101
reason_reputation    -0.0374      0.095     -0.396      0.693      -0.223
0.148
guardian_mother      -0.0706      0.085     -0.833      0.405      -0.237
0.096
guardian_other       -0.0310      0.169     -0.183      0.855      -0.363
0.301
==============================================================================
Omnibus:                       68.163   Durbin-Watson:                   1.767
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              312.974
Skew:                           0.353   Prob(JB):                     1.09e-68
Kurtosis:                       6.372   Cond. No.                         498.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
```

specified.


Residuals vs Fitted Values
Residuals
Fitted Values
4
2
0
−2
6
8
10
12
14
16
18


QQ Plot of Residuals
Sample Quantiles
Theoretical Quantiles
6
4
2
0
−2
−4
−4
−2
0
2
4
6



Interpretation

The residuals of the numerical features appear to be normally distributed.

0.7.4 Heteroscedasticity

[53]:
```
#  --------------- Test for Heteroscedasticity (Breusch-Pagan) ------------------

# Residuals and predictors from preliminary OLS model
lm = prelim_model  # your OLS model

# Run Breusch-Pagan test
bp_test = het_breuschpagan(lm.resid, lm.model.exog)
bp_labels = ['LM Statistic', 'LM p-value', 'F-Statistic', 'F p-value']

print("Breusch-Pagan Test for Heteroscedasticity:")
for label, value in zip(bp_labels, bp_test):
    print(f"{label}: {value:.4f}")
```

```
Breusch-Pagan Test for Heteroscedasticity:
LM Statistic: 67.3687
LM p-value: 0.0058
F-Statistic: 1.7168
F p-value: 0.0043
```

Observation

The test indicates significant heteroscedasticity, as evidenced by p-values near zero, leading to rejection of the null hypothesis. Consequently, OLS estimates may have inefficient standard errors; it is recommended to use robust

standard errors and consider alternative modeling approaches that are less sensitive to heteroscedasticity, such as tree-based methods like Random Forest or Gradient Boosting.

0.8 Modeling

0.8.1 Feature Engineering

We will test an OLS model with few interaction terms to see if we should include them in our final model:

[54]:
```
# ------------------- Feature Engineering & Interaction Terms -------------------
X_test_fe = df_encoded.drop(columns=['G3']).copy()
y_test_fe = df_no_outliers['G3']
X_test_fe = X_test_fe.loc[y_test_fe.index]

# 1. Performance improvement (abs difference between G1 and G2)
X_test_fe['G_perf_improve'] = abs(df_no_outliers['G2'] - df_no_outliers['G1'])

# 2. Average past performance (G1 + G2)/2
X_test_fe['G_avg'] = (df_no_outliers['G1'] + df_no_outliers['G2']) / 2

# 3. Interaction term (studytime * failures)
if 'studytime' in X_test_fe.columns and 'failures' in X_test_fe.columns:
    X_test_fe['studytime_failures'] = X_test_fe['studytime'] * X_test_fe['failures']

# 4. Interaction term (absences * studytime)
if 'absences' in X_test_fe.columns and 'studytime' in X_test_fe.columns:
    X_test_fe['absences_studytime'] = X_test_fe['absences'] * X_test_fe['studytime']

# ------------------- Train/Test Split -------------------
X_train_fe, X_test_fe, y_train_fe, y_test_fe = train_test_split(
    X_test_fe, y_test_fe, test_size=0.2, random_state=42
)

# ------------------- OLS: Baseline -------------------
X_sm_base = sm.add_constant(X_train_fe.
 ↪drop(columns=['G_perf_improve','studytime_failures','G_avg','absences_studytime']))
ols_base = sm.OLS(y_train_fe, X_sm_base).fit()
y_pred_base = ols_base.predict(sm.add_constant(X_test_fe.
 ↪drop(columns=['G_perf_improve','studytime_failures','G_avg','absences_studytime'])))

print("Baseline OLS Test R2:", r2_score(y_test_fe, y_pred_base))
print("Baseline OLS Test RMSE:", mean_squared_error(y_test_fe, y_pred_base) ** 0.5)

# ------------------- OLS: With Feature-Engineered Terms -------------------
X_sm_fe = sm.add_constant(X_train_fe)
ols_fe = sm.OLS(y_train_fe, X_sm_fe).fit()
y_pred_fe = ols_fe.predict(sm.add_constant(X_test_fe))

print("FE OLS Test R2:", r2_score(y_test_fe, y_pred_fe))
print("FE OLS Test RMSE:", mean_squared_error(y_test_fe, y_pred_fe) ** 0.5)
```

```
Baseline OLS Test R2: 0.8949719417225516
Baseline OLS Test RMSE: 0.8717123735429568
FE OLS Test R2: 0.8940081828334319
FE OLS Test RMSE: 0.8757027456172547
```

Observation

The test set results show no improvement after adding the feature-engineered and interaction terms. R^2 and RMSE

are largely unchanged, indicating these features do not add meaningful predictive value.

0.8.2 Comparing Models

We will use all predictors to choose the best model between OLS, Lasso, Ridge, Random Forest, and Gradient Boosting:

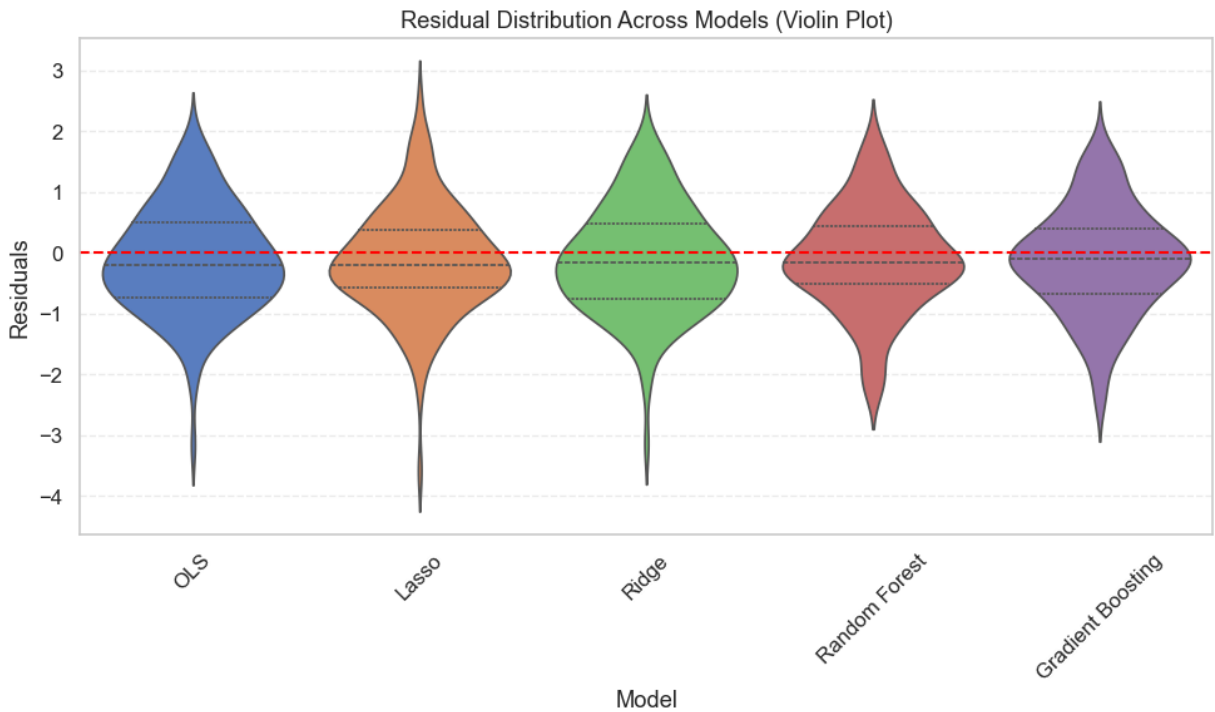
[55]:
```
# ------------------- Data -------------------
X = df_encoded.drop(columns=['G3'])
y = df_no_outliers['G3']  # keep outlier removal
X = X.loc[y.index]

# Train/test split with random seed for reproducibility
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Initialize results container
results = []

# ------------------- 1. Scaled OLS -------------------

# Scale features
scaler = StandardScaler()
X_train_scaled = pd.DataFrame(scaler.fit_transform(X_train), columns=X_train.columns,␣
↪index=X_train.index)
X_test_scaled  = pd.DataFrame(scaler.transform(X_test), columns=X_test.columns, index=X_test.
↪index)

# Add constant for intercept
X_train_scaled = sm.add_constant(X_train_scaled)
X_test_scaled  = sm.add_constant(X_test_scaled)

# Fit OLS on scaled features
ols_model = sm.OLS(y_train, X_train_scaled).fit(cov_type='HC3') # robust SEs

# Predictions
y_pred_train_ols = ols_model.predict(X_train_scaled)
y_pred_test_ols  = ols_model.predict(X_test_scaled)

results.append({
    "Model": "OLS",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_ols) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_ols) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_ols),
    "Test_R2": r2_score(y_test, y_pred_test_ols)
})

# ------------------- 2. Lasso -------------------
lasso_pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('lasso', Lasso(random_state=42))
])

param_grid = {'lasso__alpha': [0.001, 0.01, 0.1, 1, 10]}
grid_lasso = GridSearchCV(lasso_pipe, param_grid, cv=5, scoring='r2')
grid_lasso.fit(X_train, y_train)
best_lasso = grid_lasso.best_estimator_
```

```python
y_pred_train_lasso = best_lasso.predict(X_train)
y_pred_test_lasso  = best_lasso.predict(X_test)

results.append({
    "Model": "Lasso",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_lasso) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_lasso) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_lasso),
    "Test_R2": r2_score(y_test, y_pred_test_lasso)
})

# ------------------- 3. Ridge -------------------
ridge_pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('ridge', Ridge(random_state=42))
])

param_grid = {'ridge__alpha': [0.01, 0.1, 1, 10, 100]}
grid_ridge = GridSearchCV(ridge_pipe, param_grid, cv=5, scoring='r2')
grid_ridge.fit(X_train, y_train)
best_ridge = grid_ridge.best_estimator_

y_pred_train_ridge = best_ridge.predict(X_train)
y_pred_test_ridge  = best_ridge.predict(X_test)

results.append({
    "Model": "Ridge",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_ridge) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_ridge) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_ridge),
    "Test_R2": r2_score(y_test, y_pred_test_ridge)
})

# ------------------- 4. Random Forest -------------------
rf = RandomForestRegressor(random_state=42, n_estimators=200)
rf.fit(X_train, y_train)

y_pred_train_rf = rf.predict(X_train)
y_pred_test_rf  = rf.predict(X_test)

results.append({
    "Model": "Random Forest",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_rf) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_rf) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_rf),
    "Test_R2": r2_score(y_test, y_pred_test_rf)
})

# ------------------- 5. Gradient Boosting -------------------
gb = GradientBoostingRegressor(random_state=42, n_estimators=200)
gb.fit(X_train, y_train)

y_pred_train_gb = gb.predict(X_train)
y_pred_test_gb  = gb.predict(X_test)

results.append({
```

```
    "Model": "Gradient Boosting",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_gb) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_gb) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_gb),
    "Test_R2": r2_score(y_test, y_pred_test_gb)
})

# ------------------- Convert results to DataFrame -------------------
results_df = pd.DataFrame(results)
display(results_df)

# ------------------- Plot comparisons -------------------

# Determine the best model (lowest Test RMSE)
best_idx = results_df['Test_RMSE'].idxmin()

# Bar width and positions
x = np.arange(len(results_df['Model']))
width = 0.35

fig, axes = plt.subplots(1, 2, figsize=(12,6))

# Colors
train_color = 'skyblue'
test_color = 'salmon'
highlight_edge = 'black'

# RMSE Comparison
for i in range(len(x)):
    edge = highlight_edge if i == best_idx else None
    axes[0].bar(x[i] - width/2, results_df['Train_RMSE'][i], width=width, color=train_color,␣
 ↪edgecolor=edge, linewidth=2 if edge else 1, label='Train RMSE' if i==0 else "")
    axes[0].bar(x[i] + width/2, results_df['Test_RMSE'][i], width=width, color=test_color,␣
 ↪edgecolor=edge, linewidth=2 if edge else 1, label='Test RMSE' if i==0 else "")

axes[0].set_xticks(x)
axes[0].set_xticklabels(results_df['Model'], rotation=45, ha='right')
axes[0].set_title("RMSE Comparison")
axes[0].set_ylabel("RMSE")
axes[0].grid(axis='y', linestyle='--', alpha=0.5)
axes[0].legend(loc='lower right')

# R² Comparison
for i in range(len(x)):
    edge = highlight_edge if i == best_idx else None
    axes[1].bar(x[i] - width/2, results_df['Train_R2'][i], width=width, color=train_color,␣
 ↪edgecolor=edge, linewidth=2 if edge else 1, label='Train R²' if i==0 else "")
    axes[1].bar(x[i] + width/2, results_df['Test_R2'][i], width=width, color=test_color,␣
 ↪edgecolor=edge, linewidth=2 if edge else 1, label='Test R²' if i==0 else "")

axes[1].set_xticks(x)
axes[1].set_xticklabels(results_df['Model'], rotation=45, ha='right')
axes[1].set_title("R² Comparison")
axes[1].set_ylabel("R²")
axes[1].grid(axis='y', linestyle='--', alpha=0.5)
axes[1].legend(loc='lower right')
```

```
plt.suptitle("Model Performance Comparison", fontsize=16, weight='bold')
plt.tight_layout(rect=[0,0,1,0.95])
plt.savefig("../visualizations/model_performance_comparison.png", dpi=300, bbox_inches='tight')
plt.show()

# ---------------- Violin Plot ----------------
residuals_dict = {
    "OLS": y_test - y_pred_test_ols,
    "Lasso": y_test - y_pred_test_lasso,
    "Ridge": y_test - y_pred_test_ridge,
    "Random Forest": y_test - y_pred_test_rf,
    "Gradient Boosting": y_test - y_pred_test_gb
}

plt.figure(figsize=(10,6))
sns.violinplot(
    data=pd.DataFrame(residuals_dict),
    palette="muted",
    inner="quartile"
)
plt.axhline(0, color='red', linestyle='--')
plt.title("Residual Distribution Across Models (Violin Plot)")
plt.ylabel("Residuals")
plt.xlabel("Model")
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.4)
plt.tight_layout()
plt.savefig("../visualizations/residuals_violin_plot.png", dpi=300, bbox_inches='tight')
plt.show()
```

	Model	Train_RMSE	Test_RMSE	Train_R2	Test_R2
0	OLS	0.800243	0.871712	0.908518	0.894972
1	Lasso	0.848957	0.890324	0.897041	0.890439
2	Ridge	0.802309	0.877693	0.908045	0.893526
3	Random Forest	0.334208	0.861520	0.984044	0.897414
4	Gradient Boosting	0.425179	0.868352	0.974175	0.895780



Explanation

We selected the Random Forest as the best-performing model because it achieved the lowest test set RMSE. The violin plots show that its residuals are tightly clustered around zero, with a narrow, symmetric shape and the shortest range among models, indicating accurate predictions with low variance.

A test R^2 of 0.897 indicates that approximately 89.7% of the variance in the target variable is explained by the model on unseen data. This suggests a very strong predictive performance, meaning the model captures most of the systematic patterns in the data, with only about 10% of the variance left unexplained.

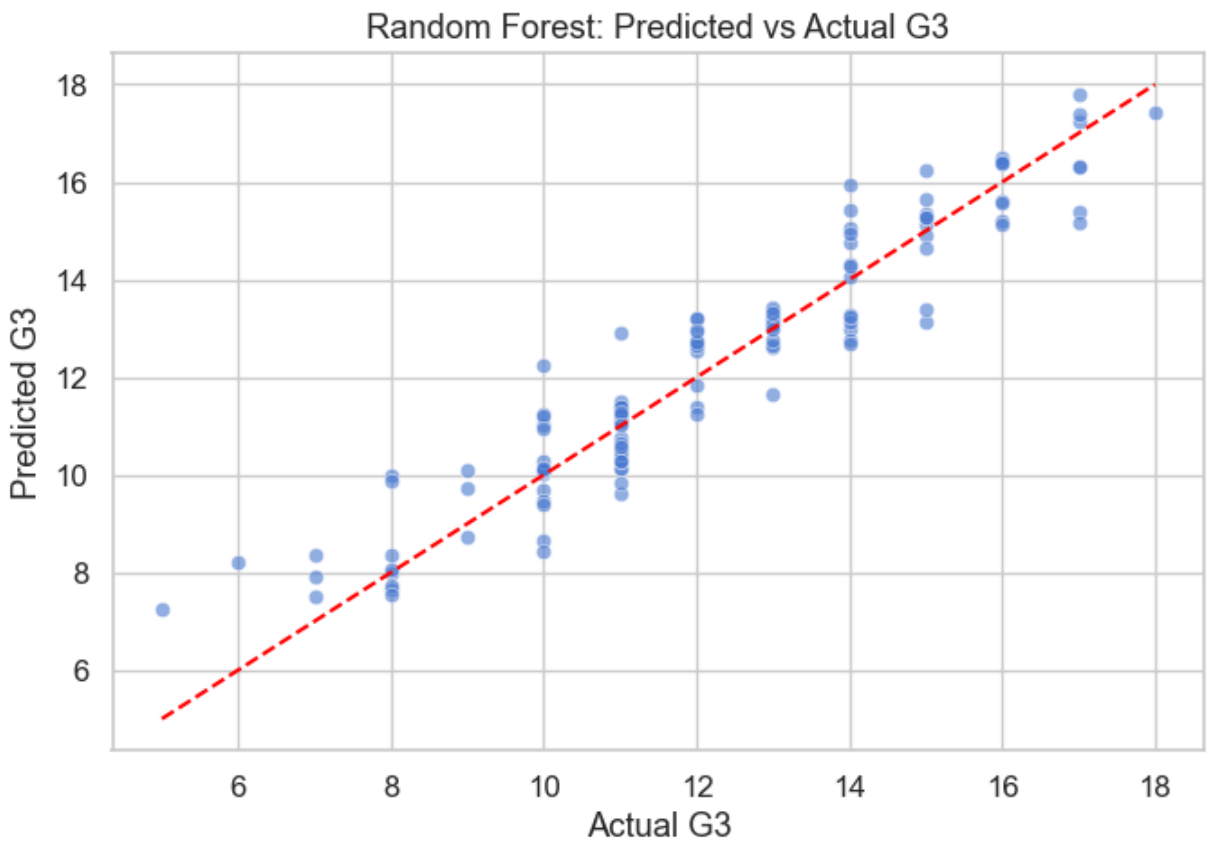
A test RMSE of 0.861 indicates that, on average, the model's predictions deviate from the true values by about 0.86 units. Combined with a high R^2, this shows that the model predicts the target variable accurately, with relatively small errors.

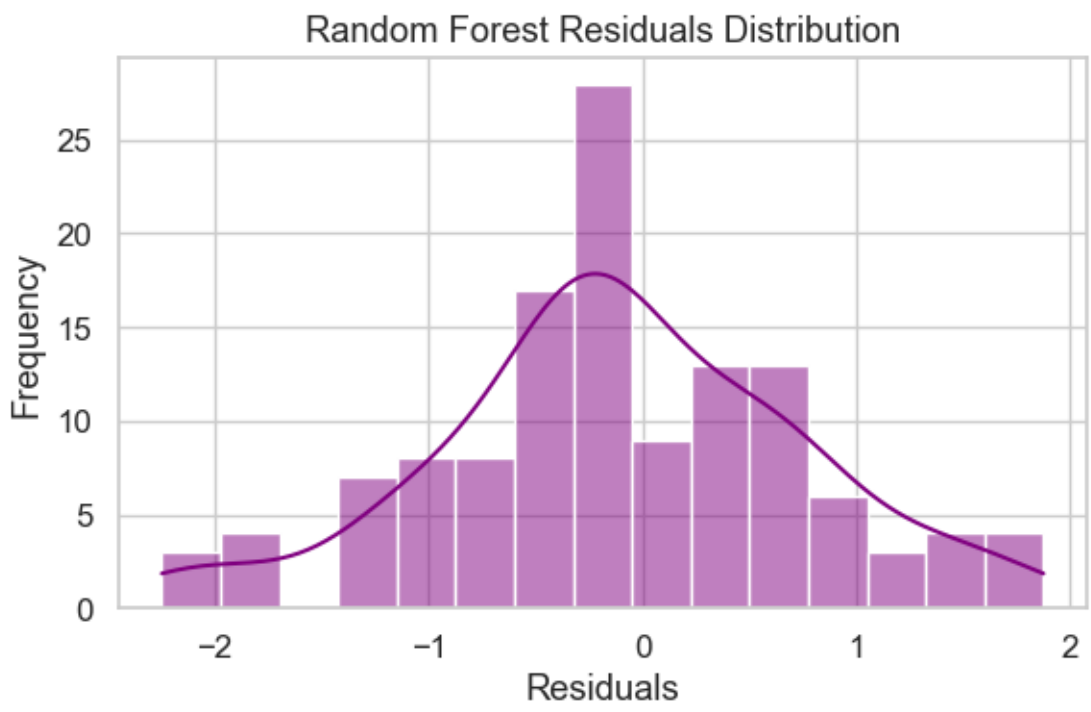
0.8.3 Interpreting the Best Model (Random Forest)

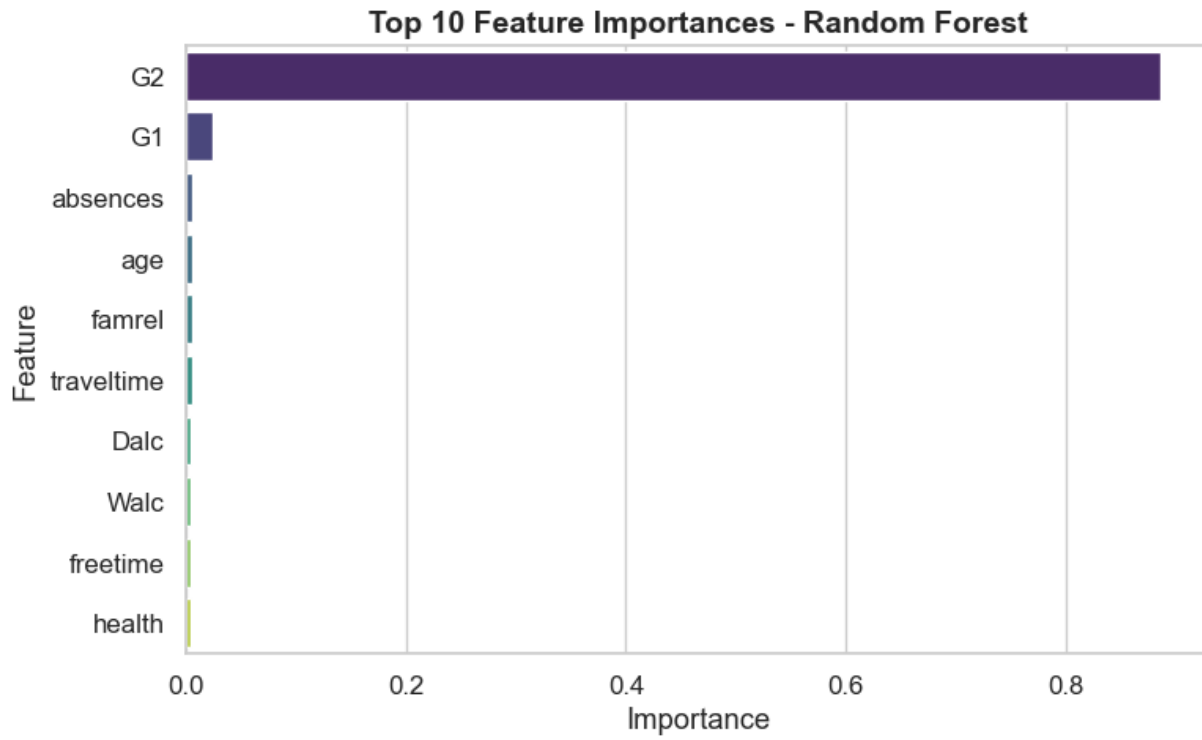
[56]:
```
# ------------------ Random Forest Interpretation ------------------

# Identify the best model index
best_idx = results_df['Test_RMSE'].idxmin()
best_model_name = results_df.loc[best_idx, 'Model']
print(f"Best model selected based on Test RMSE: {best_model_name}")

# ------------------ Predictions vs Actuals ------------------
y_pred_rf = rf.predict(X_test)

plt.figure(figsize=(7,5))
sns.scatterplot(x=y_test, y=y_pred_rf, alpha=0.6)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color='red',␣
↪linestyle='--')
plt.xlabel("Actual G3")
```

```
plt.ylabel("Predicted G3")
plt.title("Random Forest: Predicted vs Actual G3")
plt.tight_layout()
plt.savefig("../visualizations/rf_predicted_vs_actual.png", dpi=300, bbox_inches='tight')
plt.show()

# ------------------ Residuals ------------------
residuals_rf = y_test - y_pred_rf

plt.figure(figsize=(6,4))
sns.histplot(residuals_rf, kde=True, color='purple', bins=15)
plt.title("Random Forest Residuals Distribution")
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.tight_layout()
plt.savefig("../visualizations/rf_residuals_distribution.png", dpi=300, bbox_inches='tight')
plt.show()

# ------------------ Top 10 Feature Importances (Random Forest) ------------------
if best_model_name == "Random Forest":
    rf_importances = rf.feature_importances_
    feat_importance = pd.Series(rf_importances, index=X.columns).sort_values(ascending=False)

    top_10_features = feat_importance.head(10)

    plt.figure(figsize=(8,5))
    sns.barplot(
    x=top_10_features.values,
    y=top_10_features.index,
    hue=top_10_features.index,
    palette="viridis",
    legend=False
    )
    plt.title("Top 10 Feature Importances - Random Forest", fontsize=14, weight='bold')
    plt.xlabel("Importance")
    plt.ylabel("Feature")
    plt.tight_layout()
    plt.savefig("../visualizations/rf_top_10_feature_importances.png", dpi=300,␣
↪bbox_inches='tight')
    plt.show()

    print("Top 10 important features:")
    display(top_10_features)
```

```
Best model selected based on Test RMSE: Random Forest
```


Random Forest: Predicted vs Actual G3
Predicted G3
Actual G3
18
16
14
12
10
8
6
6
8
10
12
14
16
18


Random Forest Residuals Distribution
Frequency
Residuals
25
20
15
10
5
0
−2
−1
0
1
2




```
Top 10 important features:

G2            0.886439
G1            0.024254
absences      0.006254
age           0.005706
famrel        0.005571
traveltime    0.005500
Dalc          0.004274
Walc          0.004050
freetime      0.004038
health        0.004022
dtype: float64
```

Interpretation

The model's residuals are approximately normal and homoscedastic. The strongest predictors of G3 are G2, G1, absences, and age.

0.8.4 Comparing Models (again) `Without` 'G1' and 'G2' as Predictors

We will remove 'G1' and 'G2' and choose the best model between OLS, Lasso, Ridge, Random Forest, and Gradient Boosting:

[57]:
```
# ------------------- Data (remove G1 and G2) -------------------
X = df_encoded.drop(columns=['G3', 'G1', 'G2'])
y = df_no_outliers['G3']  # keep outlier removal
X = X.loc[y.index]

# Train/test split with random seed for reproducibility
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

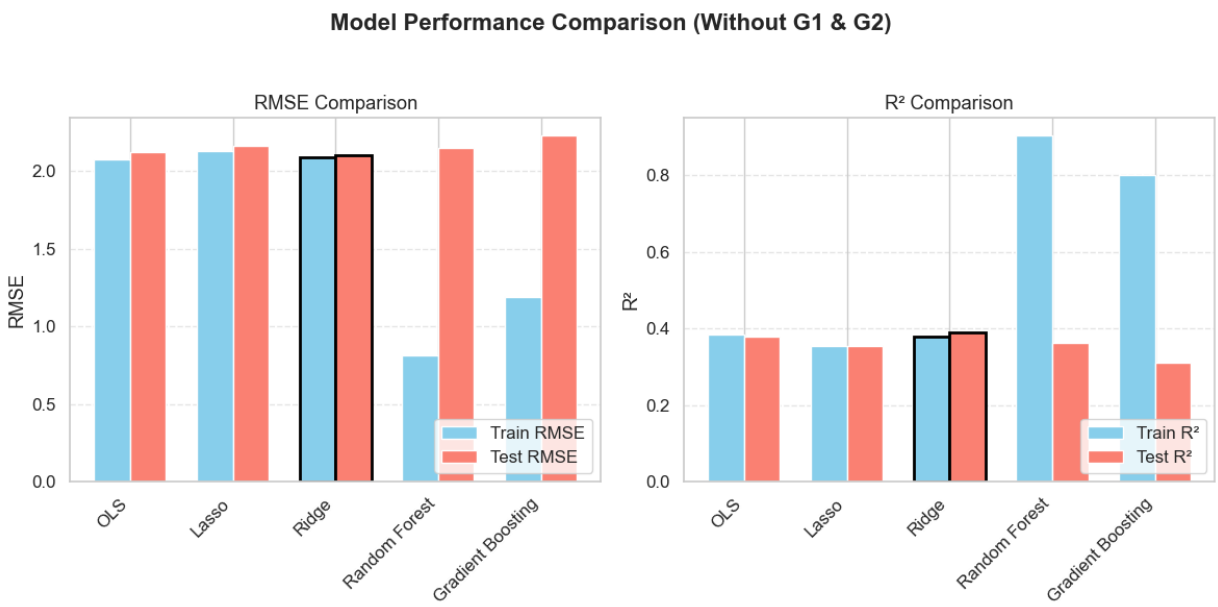
```
)

# Initialize results container
results = []

# ------------------- 1. Scaled OLS -------------------

# Scale features
scaler = StandardScaler()
X_train_scaled = pd.DataFrame(scaler.fit_transform(X_train), columns=X_train.columns,␣
↪index=X_train.index)
X_test_scaled  = pd.DataFrame(scaler.transform(X_test), columns=X_test.columns, index=X_test.
↪index)

# Add constant for intercept
X_train_scaled = sm.add_constant(X_train_scaled)
X_test_scaled  = sm.add_constant(X_test_scaled)

# Fit OLS on scaled features
ols_model = sm.OLS(y_train, X_train_scaled).fit(cov_type='HC3') # robust SEs

# Predictions
y_pred_train_ols = ols_model.predict(X_train_scaled)
y_pred_test_ols  = ols_model.predict(X_test_scaled)

results.append({
    "Model": "OLS",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_ols) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_ols) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_ols),
    "Test_R2": r2_score(y_test, y_pred_test_ols)
})

# ------------------- 2. Lasso -------------------
lasso_pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('lasso', Lasso(random_state=42))
])

param_grid = {'lasso__alpha': [0.001, 0.01, 0.1, 1, 10]}
grid_lasso = GridSearchCV(lasso_pipe, param_grid, cv=5, scoring='r2')
grid_lasso.fit(X_train, y_train)
best_lasso = grid_lasso.best_estimator_

y_pred_train_lasso = best_lasso.predict(X_train)
y_pred_test_lasso  = best_lasso.predict(X_test)

results.append({
    "Model": "Lasso",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_lasso) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_lasso) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_lasso),
    "Test_R2": r2_score(y_test, y_pred_test_lasso)
})

# ------------------- 3. Ridge -------------------
ridge_pipe = Pipeline([
```

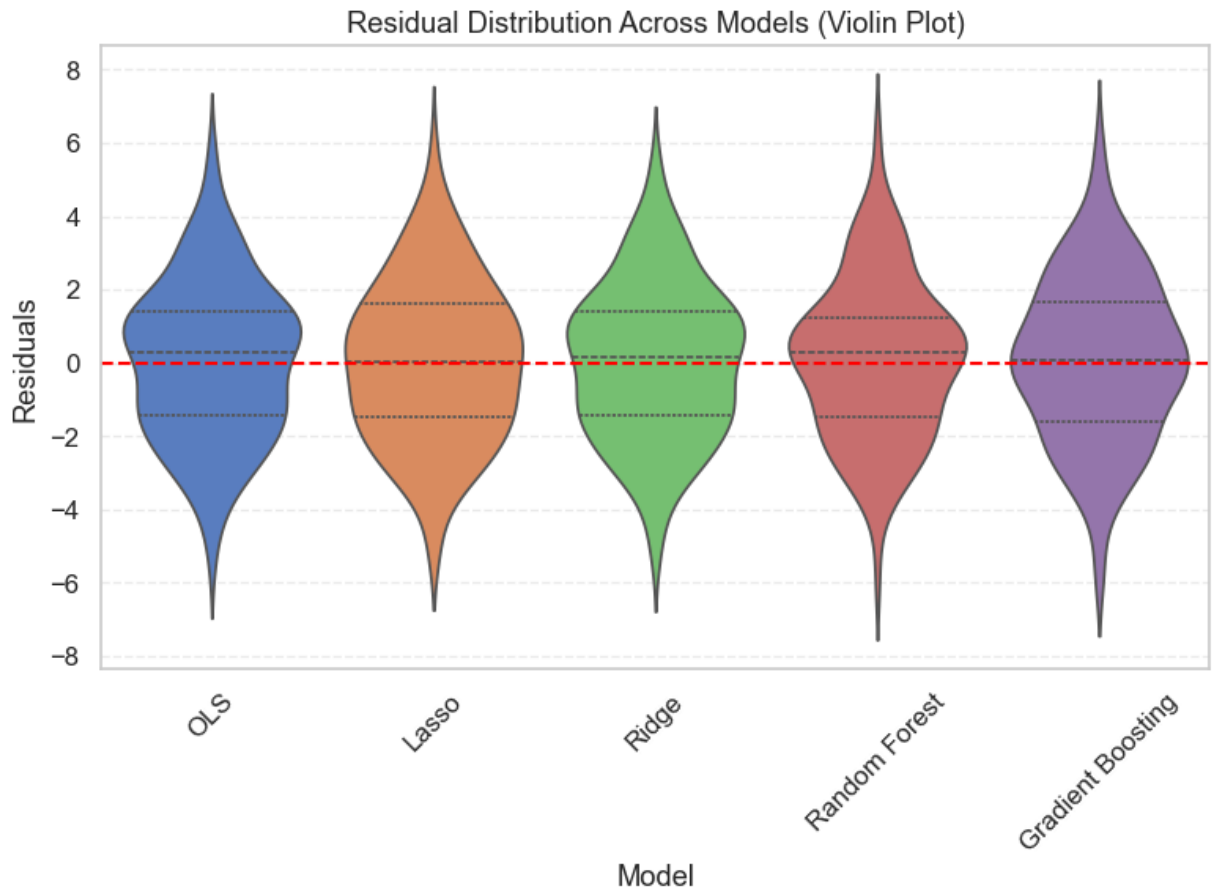
```
    ('scaler', StandardScaler()),
    ('ridge', Ridge(random_state=42))
])

param_grid = {'ridge__alpha': [0.01, 0.1, 1, 10, 100]}
grid_ridge = GridSearchCV(ridge_pipe, param_grid, cv=5, scoring='r2')
grid_ridge.fit(X_train, y_train)
best_ridge = grid_ridge.best_estimator_

y_pred_train_ridge = best_ridge.predict(X_train)
y_pred_test_ridge  = best_ridge.predict(X_test)

results.append({
    "Model": "Ridge",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_ridge) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_ridge) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_ridge),
    "Test_R2": r2_score(y_test, y_pred_test_ridge)
})

# ------------------- 4. Random Forest -------------------
rf = RandomForestRegressor(random_state=42, n_estimators=200)
rf.fit(X_train, y_train)

y_pred_train_rf = rf.predict(X_train)
y_pred_test_rf  = rf.predict(X_test)

results.append({
    "Model": "Random Forest",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_rf) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_rf) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_rf),
    "Test_R2": r2_score(y_test, y_pred_test_rf)
})

# ------------------- 5. Gradient Boosting -------------------
gb = GradientBoostingRegressor(random_state=42, n_estimators=200)
gb.fit(X_train, y_train)

y_pred_train_gb = gb.predict(X_train)
y_pred_test_gb  = gb.predict(X_test)

results.append({
    "Model": "Gradient Boosting",
    "Train_RMSE": mean_squared_error(y_train, y_pred_train_gb) ** 0.5,
    "Test_RMSE": mean_squared_error(y_test, y_pred_test_gb) ** 0.5,
    "Train_R2": r2_score(y_train, y_pred_train_gb),
    "Test_R2": r2_score(y_test, y_pred_test_gb)
})

# ------------------- Convert results to DataFrame -------------------
results_df = pd.DataFrame(results)
display(results_df)

# ------------------- Plot comparisons -------------------
best_idx = results_df['Test_RMSE'].idxmin()
x = np.arange(len(results_df['Model']))
```

```
width = 0.35

fig, axes = plt.subplots(1, 2, figsize=(12,6))
train_color = 'skyblue'
test_color = 'salmon'
highlight_edge = 'black'

# RMSE
for i in range(len(x)):
    edge = highlight_edge if i == best_idx else None
    axes[0].bar(x[i] - width/2, results_df['Train_RMSE'][i], width=width, color=train_color,
                edgecolor=edge, linewidth=2 if edge else 1, label='Train RMSE' if i==0 else "")
    axes[0].bar(x[i] + width/2, results_df['Test_RMSE'][i], width=width, color=test_color,
                edgecolor=edge, linewidth=2 if edge else 1, label='Test RMSE' if i==0 else "")

axes[0].set_xticks(x)
axes[0].set_xticklabels(results_df['Model'], rotation=45, ha='right')
axes[0].set_title("RMSE Comparison")
axes[0].set_ylabel("RMSE")
axes[0].grid(axis='y', linestyle='--', alpha=0.5)
axes[0].legend(loc='lower right')

# R²
for i in range(len(x)):
    edge = highlight_edge if i == best_idx else None
    axes[1].bar(x[i] - width/2, results_df['Train_R2'][i], width=width, color=train_color,
                edgecolor=edge, linewidth=2 if edge else 1, label='Train R²' if i==0 else "")
    axes[1].bar(x[i] + width/2, results_df['Test_R2'][i], width=width, color=test_color,
                edgecolor=edge, linewidth=2 if edge else 1, label='Test R²' if i==0 else "")

axes[1].set_xticks(x)
axes[1].set_xticklabels(results_df['Model'], rotation=45, ha='right')
axes[1].set_title("R² Comparison")
axes[1].set_ylabel("R²")
axes[1].grid(axis='y', linestyle='--', alpha=0.5)
axes[1].legend(loc='lower right')

plt.suptitle("Model Performance Comparison (Without G1 & G2)", fontsize=16, weight='bold')
plt.tight_layout(rect=[0,0,1,0.95])
plt.savefig("../visualizations/model_performance_comparison_no_g1_g2.png", dpi=300,␣
↪bbox_inches='tight')
plt.show()

# ---------------- Violin Plot ----------------
residuals_dict = {
    "OLS": y_test - y_pred_test_ols,
    "Lasso": y_test - y_pred_test_lasso,
    "Ridge": y_test - y_pred_test_ridge,
    "Random Forest": y_test - y_pred_test_rf,
    "Gradient Boosting": y_test - y_pred_test_gb
}

plt.figure(figsize=(8,6))
sns.violinplot(
    data=pd.DataFrame(residuals_dict),  # use wide-form directly
    palette="muted",
    inner="quartile"
```

```
)
plt.axhline(0, color='red', linestyle='--')
plt.title("Residual Distribution Across Models (Violin Plot)")
plt.ylabel("Residuals")
plt.xlabel("Model")
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.4)
plt.tight_layout()
plt.savefig("../visualizations/residuals_violin_plot_no_g1_g2.png", dpi=300,␣
 ↪bbox_inches='tight')
plt.show()
```

```
               Model  Train_RMSE  Test_RMSE  Train_R2   Test_R2
0                OLS    2.076106   2.122731  0.384270  0.377200
1              Lasso    2.128481   2.161344  0.352812  0.354336
2              Ridge    2.087657   2.101231  0.377399  0.389752
3      Random Forest    0.815488   2.146161  0.904999  0.363375
4  Gradient Boosting    1.186433   2.231479  0.798916  0.311753
```





Explanation

We selected Ridge regression as the best-performing model because it achieved the lowest test set RMSE. The violin plots show that its residuals are tightly clustered around zero, with short tails indicating a small range of errors. Although the zero line runs through the center of the Lasso residuals as well, Ridge was preferred due to its slightly lower RMSE, reflected in the narrower distribution of residuals.

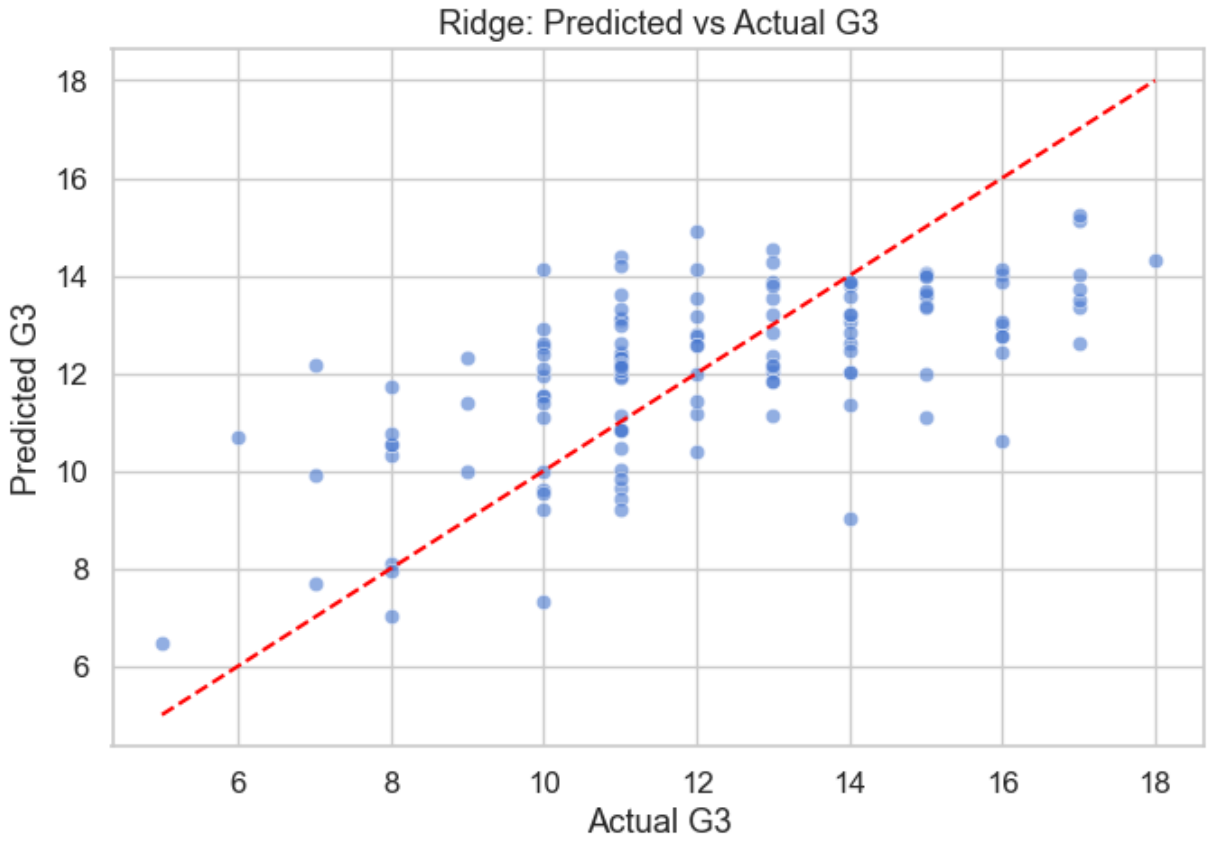
As expected, the predictive power of this model with G1 and G2 removed is much lower (r2 = .39), although the purpose is to analyze how the other feature importances will change without the G1 and G2 predictors. We will do this next.

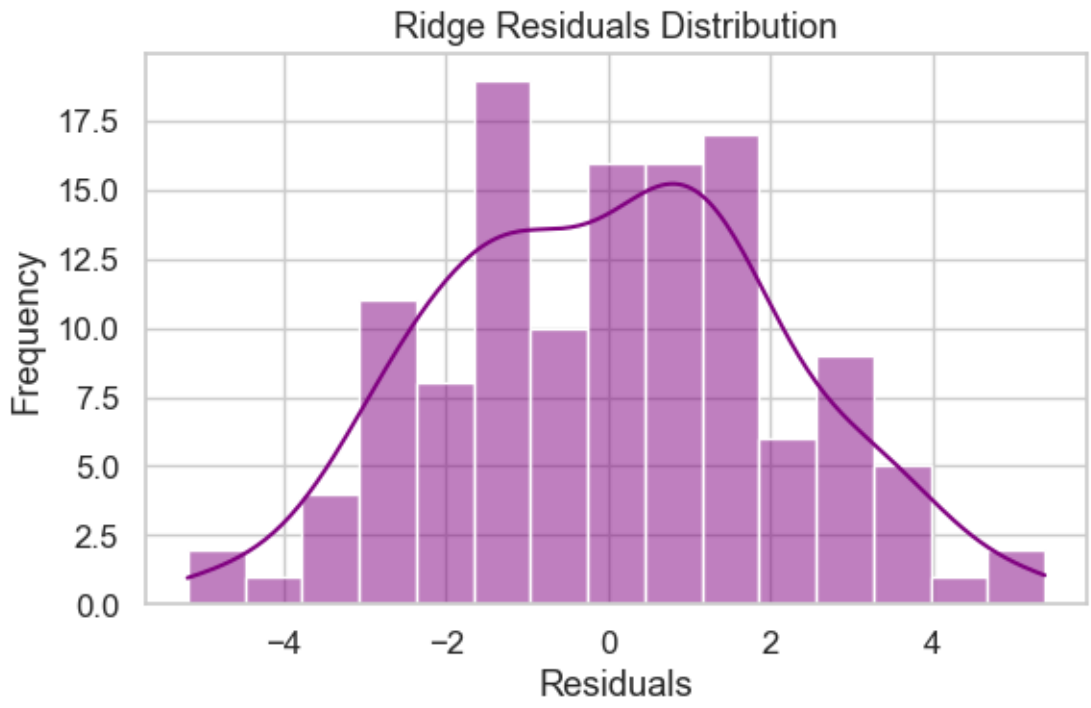
0.8.5 Interpreting Best Model (again) `Without` 'G1' and 'G2' as Predictors (Ridge)

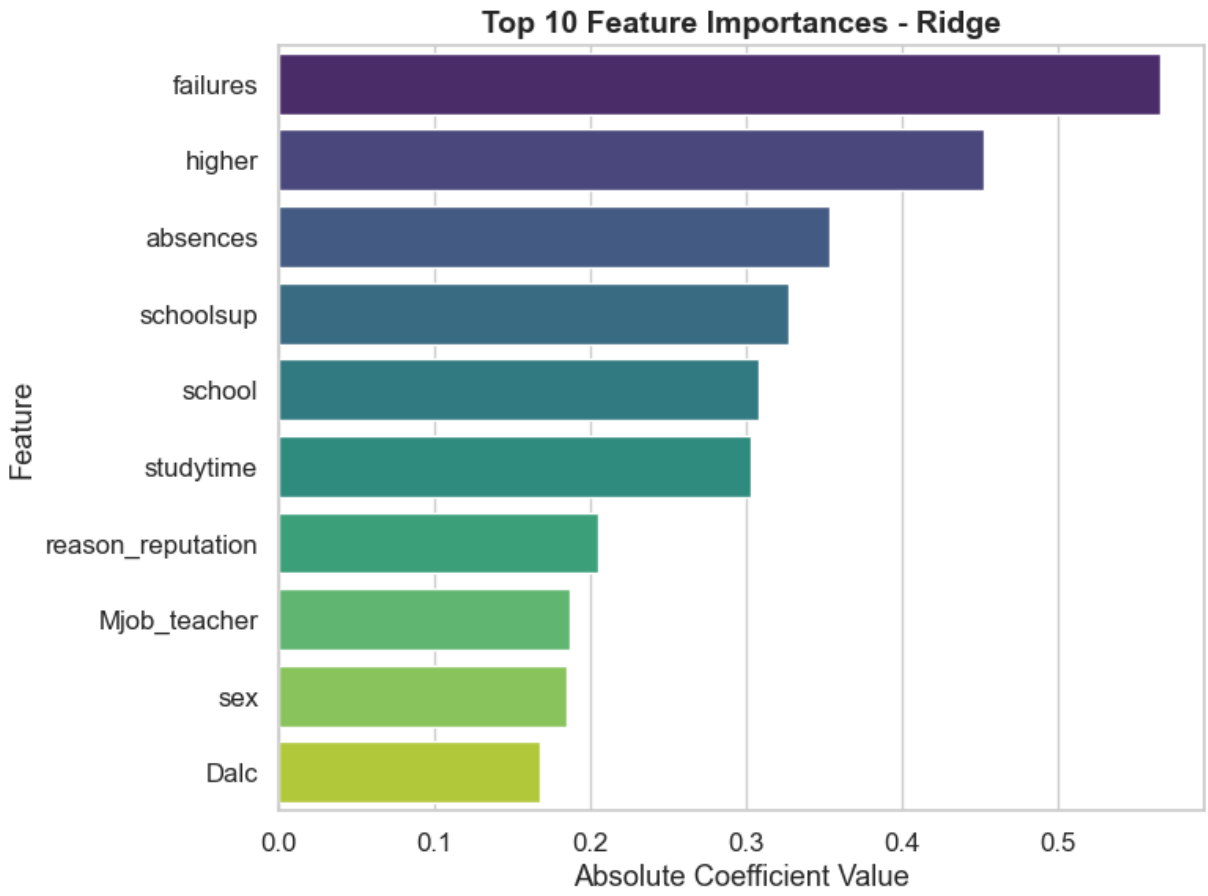
[58]:
```
# ------------------ Ridge Model Interpretation ------------------

# Identify the best model index
best_idx = results_df['Test_RMSE'].idxmin()
best_model_name = results_df.loc[best_idx, 'Model']
print(f"Best model selected based on Test RMSE: {best_model_name}")

# ------------------ Predictions vs Actuals ------------------
# Use Ridge predictions
y_pred_ridge = best_ridge.predict(X_test)

plt.figure(figsize=(7,5))
sns.scatterplot(x=y_test, y=y_pred_ridge, alpha=0.6)
```

```
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color='red',␣
↪linestyle='--')
plt.xlabel("Actual G3")
plt.ylabel("Predicted G3")
plt.title("Ridge: Predicted vs Actual G3")
plt.tight_layout()
plt.savefig("../visualizations/ridge_predicted_vs_actual.png", dpi=300, bbox_inches='tight')
plt.show()

# ------------------ Residuals ------------------
residuals_ridge = y_test - y_pred_ridge

plt.figure(figsize=(6,4))
sns.histplot(residuals_ridge, kde=True, color='purple', bins=15)
plt.title("Ridge Residuals Distribution")
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.tight_layout()
plt.savefig("../visualizations/ridge_residuals_distribution.png", dpi=300, bbox_inches='tight')
plt.show()

# ------------------ Top 10 Feature Importances (Ridge) ------------------
if best_model_name == "Ridge":
    ridge_coefs = best_ridge.named_steps['ridge'].coef_
    feat_importance = pd.Series(ridge_coefs, index=X.columns).abs().sort_values(ascending=False)

    top_10_features = feat_importance.head(10)

    plt.figure(figsize=(8,6))
    sns.barplot(
    x=top_10_features.values,
    y=top_10_features.index,
    hue=top_10_features.index,
    palette="viridis",
    legend=False
    )
    plt.title("Top 10 Feature Importances - Ridge", fontsize=14, weight='bold')
    plt.xlabel("Absolute Coefficient Value")
    plt.ylabel("Feature")
    plt.tight_layout()
    plt.savefig("../visualizations/ridge_top_10_feature_importances.png", dpi=300,␣
↪bbox_inches='tight')
    plt.show()

    print("Top 10 important features:")
    display(top_10_features)
```

```
Best model selected based on Test RMSE: Ridge
```


Ridge: Predicted vs Actual G3
Predicted G3
Actual G3
18
16
14
12
10
8
6
6
8
10
12
14
16
18


Ridge Residuals Distribution
Frequency
Residuals
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−4
−2
0
2
4




```
Top 10 important features:

failures             0.565870
higher               0.452681
absences             0.353691
schoolsup            0.326584
school               0.307403
studytime            0.302993
reason_reputation    0.204352
Mjob_teacher         0.186747
sex                  0.184888
Dalc                 0.167625
dtype: float64
```

Interpretation:

The model's residuals are approximately normal and homoscedastic. The strongest predictors in this model with G1 and G2 removed are now failuers, higher (pursuit of higher education), absences, schoolsup (level of academic support), and school.